Localized Matched-Subspace Detection for Functional Ultrasound and Ultrafast Doppler Imaging

Arthur Shune*
SkyMesa Systems Inc.

Abstract

Functional ultrasound is now used for intraoperative, neonatal, and mobile brain imaging, but even after standard clutter suppression, tissue motion, phase instability, and aliased flow can produce false detections that distort vascular maps and contaminate functional readouts. We study a family of localized matched-subspace detector heads applied after conventional clutter suppression of beamformed fUS slow-time data, ranging from a fixed band-limited score to locally whitened variants. Quantitative evidence comes from labeled brain simulations, a structurally labeled flow phantom, a held-out SIMUS moving-scatterer benchmark, and limited real-IQ audits. On the held-out synthetic structural benchmark, the fixed head reduces nuisance FPR from 0.998 to 0.004 at matched recall 0.5. On one open real-IQ rat-brain dataset, only the fully whitened variant shows a modest structural gain, improving AUC from 0.504 for power Doppler to 0.530 on a same-residual localization-derived vessel-core versus shell audit. The preferred variant is regime-dependent. These results show that changing only the downstream score head, without modifying the upstream clutter filter, can materially improve map trustworthiness, while real-data functional task-activation validation remains future work.

*Correspondence: `arthur@skymesasystems.com`. ORCID: `0009-0004-6433-3856`.

Evidence at a glance. The evidence spans four tiers, from controlled labeled detection to more limited real-IQ corroboration. The validation ladder below summarizes that hierarchy before the technical details.

Benchmark label	Independent anchor	Primary use
kWave-Stress	simulator truth	controlled mechanism checks and strict-tail sensitivity under labeled nuisance stress
Gammex-Struct	B-mode structural labels	structural ROC on real phantom data with independent geometry
SIMUS-Struct-Mobile / SIMUS-Struct-Intraop	simulated label masks in use-case-motivated scenes	main held-out structural benchmark, clutter-heavy stress tests, and fixed-calibration transfer
ULM-CoreShell	published localization-derived vascular reference	structural audit on real in-vivo IQ plus block-split fixed-calibration transfer
ULM-Motion / Shin-ProxyMotion	no independent task labels	robustness evidence and secondary real-IQ fixed-threshold corroboration

Table 1: Compact validation ladder. For readability, the main text uses the short benchmark labels listed here: kWave-Stress, Gammex-Struct, SIMUS-Struct-Mobile, SIMUS-Struct-Intraop, ULM-CoreShell, ULM-Motion, and Shin-ProxyMotion. The evidence progresses from labeled simulation and structurally labeled phantom data to a held-out use-case-motivated synthetic benchmark and open real in-vivo IQ audits.

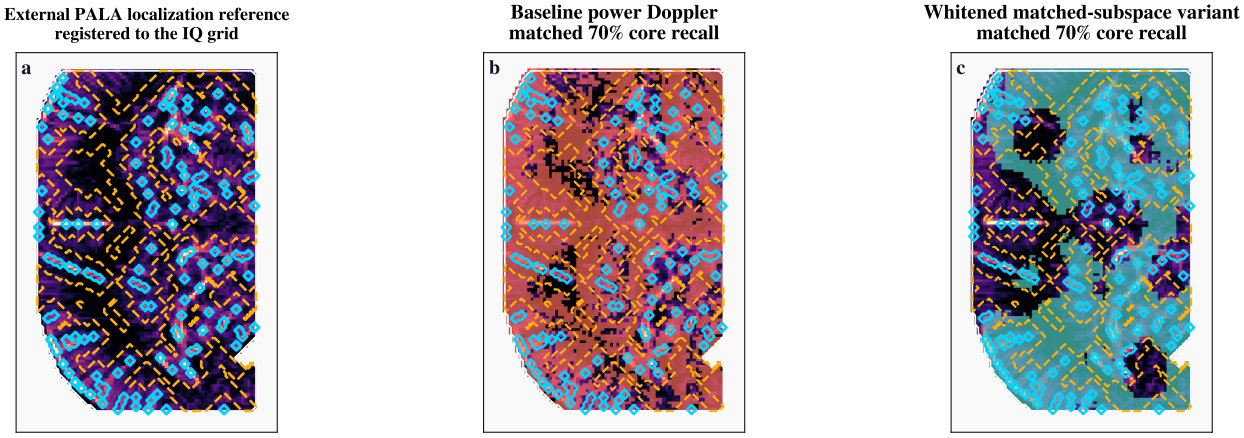

Figure 1: Localization-derived structural-label result on ULM Zenodo 7883227 using the published PALA rat-brain localization rendering as an external vascular reference. Left: the PALA-derived vessel-core and perivascular-shell masks after registration back to the IQ grid. Middle and right: the same audited 64-frame MC–SVD residual window (block 001, frames 128:192), thresholded at matched 70% vessel-core recall for baseline power Doppler and the prespecified no-registration whitened matched-subspace variant. Table 2 reports the formal same-residual head comparison.

Score	AUC (H1 vs H0)	TPR@10^{-2}	TPR@10^{-3}	shell FPR @70% core recall
Baseline (power Doppler)	0.504 [0.498,0.510]	0.010 [0.009,0.011]	7.9e-5 [0,2.4e-4]	0.698 [0.689,0.706]
Baseline (Kasai lag-1 magnitude)	0.503 [0.496,0.509]	0.009 [0.008,0.010]	5.9e-5 [0,1.8e-4]	0.701 [0.693,0.709]
Fixed matched-subspace head	0.514 [0.508,0.521]	0.014 [0.013,0.016]	2.6e-4 [1.2e-4,4.3e-4]	0.703 [0.694,0.711]
Adaptive head	0.514 [0.507,0.520]	0.014 [0.013,0.016]	2.6e-4 [1.2e-4,4.3e-4]	0.702 [0.693,0.711]
Fully whitened variant	0.530 [0.524,0.537]	0.024 [0.021,0.027]	0.004 [0.003,0.006]	0.676 [0.665,0.687]

Table 2: Same-residual ULM structural audit on real in vivo rat-brain IQ. All rows use the same frozen 64-frame MC–SVD residual cube, the same localization-derived vessel-core versus shell masks, and the same SCM whitening recipe where applicable; only the downstream score head changes. Entries report window-level means with 95% bootstrap intervals over the 70 audited windows. Against baseline power Doppler, the block-mean AUC difference for the fully whitened variant gives an exact paired sign-flip test p-value of 0.00195.

1 Introduction

Functional ultrasound (fUS) is now used for vascular and functional mapping in settings ranging from preclinical neuroscience to awake surgery and cranial-window human imaging. In these settings, the practical problem is often not average sensitivity alone. A small number of nuisance-driven false detections can distort vascular readouts, contaminate functional maps, and undermine confidence in interpretation. Tissue motion, phase instability, and aliased high-velocity flow therefore matter most when they survive ordinary clutter suppression and populate the extreme right tail of the detector output.

In intra-operative settings, a false-positive activation region can mislead surgical planning. In resting-state connectivity studies, a single artifactual hot spot that survives across blocks can create a spurious seed correlation. In neonatal bedside monitoring, where acquisition windows are short and motion is unavoidable, nuisance leakage in the score map can be hard to distinguish from genuine hemodynamic variation without independent confirmation that is rarely available. These are not average-sensitivity problems; they are rare-event problems driven by the right tail of the score distribution.

We address that problem with a localized matched-subspace detector framework for post-residual fUS. After ordinary clutter suppression, the method asks a local question: within a small neighborhood, is the remaining slow-time signal concentrated in a prespecified flow band, or is it more consistent with structured nuisance? The contribution is a detector-head family: the fixed matched-subspace head is the core and default method, the adaptive variant selectively invokes whitening, and the fully whitened variant applies whitening throughout. Among the downstream heads tested on the held-out SIMUS-Struct benchmark, a band-limited matched-subspace score without local whitening is the strongest fixed choice. That mechanism is not universal: in the short-ensemble Gammex phantom, whitening-dominant scoring is stronger than the matched-subspace ratio on the same residual. Figure 1 previews the clearest real-data result, and the same-residual table below shows the regime dependence directly.

Terminology. Throughout the paper, we refer to the three detector modes as the fixed matched-subspace detector, the adaptive detector, and the fully whitened detector. We use the full descriptive terms local slow-time whitening and empirical tail calibration rather than shorter signal-processing abbreviations.

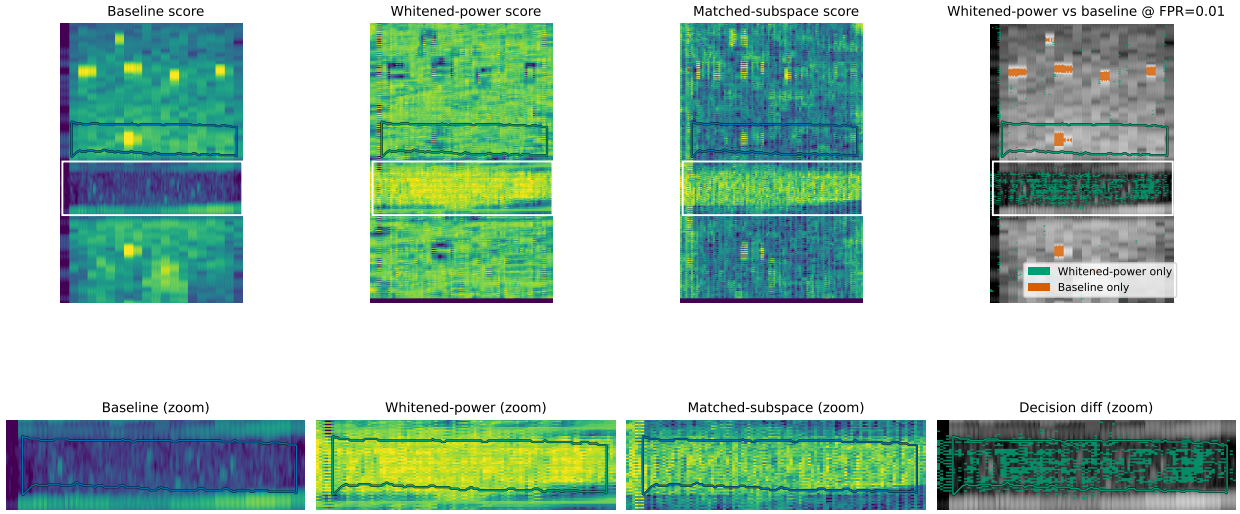

Figure 2: Leading structural-fidelity visual on a structurally labeled Gammex flow phantom acquisition from the open Doppler Twinkling Artifact database (labels derived from B-mode-only tube segmentation). Top row: baseline score map S_{base} (SVD band-pass clutter suppression followed by power Doppler), the whitened-power score map, the matched-subspace score map, and a matched-FPR decision-difference panel at a fixed background false-positive rate (10^{-2}; chosen here for visualization), showing pixels detected only by whitened power versus only by the baseline. Bottom row: zoomed region of interest. The tube contour is overlaid for reference. The qualitative ordering now matches the later Gammex structural ROC tables: whitened power is the strongest phantom score in this short-ensemble regime, while the matched-subspace panel remains included as the detector-core reference.

The effect is not confined to the extreme tail. On the representative Gammex frame in Figure 2, matching the background FPR at 10^{-2} yields zero lumen recovery for the baseline score, whereas both detector-family score maps recover lumen-consistent structure at the same matched background allowance, with the whitened-power map showing the cleaner phantom response. The later Gammex structural ROC tables confirm that whitened power is the stronger phantom score overall in this short-ensemble regime. We use the stricter 10^{-3}–10^{-4} operating points mainly as a stress test that exposes nuisance sensitivity cleanly, not because the practical relevance begins only there.

This problem matters now because fUS is no longer only a preclinical proof-of-concept modality. It is already used in intra-operative, neonatal, cranial-window, and emerging mobile settings [1, 8, 9, 20–22], where rare nuisance-driven false positives can matter more than average image quality.

The focus is downstream detector-head design after fixed residualization. The fixed non-whitened matched-subspace head is the default on the held-out synthetic structural benchmark, and the fully whitened variant helps in selected clutter-heavy or structure-concentration regimes. The validation ladder in Table 1 summarizes the evidence order before the notation and benchmark detail.

Contributions.

- A post-residual localized matched-subspace detector family for beamformed fUS slow-time

data, centered on a fixed band-limited head and extended by an adaptive whitening switch plus a fully whitened variant.

- Held-out evidence that the fixed non-whitened matched-subspace score is the best-performing fixed head among the evaluated downstream heads on the held-out SIMUS-Struct benchmark, together with clutter-heavy stress tests showing where the adaptive and fully whitened extensions help in selected regimes.
- Same-residual and same-cube comparisons that separate detector-head effects from residualization, using PD/Kasai, whitened power, and the unwhitened ratio as direct comparators.
- Limited transfer and replay-feasibility analyses, including fixed-threshold transfer on held-out synthetic and real-IQ audits and workstation replay latency.

Scope of empirical claims. The evidence is stratified by data regime. Quantitative efficacy results are reported where independent H_1/H_0 labels or an external structural reference are available. The detector family also has a fixed hierarchy throughout the paper: the fixed matched-subspace head is the default method, the adaptive variant selectively invokes whitening, and the fully whitened variant applies whitening throughout. After first definition, we use the short labels kWave-Stress, Gammex-Struct, SIMUS-Struct-Mobile, SIMUS-Struct-Intraop, ULM-CoreShell, ULM-Motion, and Shin-ProxyMotion; SIMUS-Struct denotes the paired mobile and intra-operative structural benchmark.

2 Background and Problem Formulation

Problem setup. For each spatial location (or tile-aggregated snapshot), let $x \in \mathbb{C}^{L_t}$ denote a slow-time vector. We consider

$$H_0 : \; x = c + n, \qquad H_1 : \; x = s + c + n,$$

where s is flow-consistent signal, c is structured clutter/motion, and n is residual noise. In practice, H_0 is spatially nonstationary and heavy-tailed, motivating local covariance adaptation and tail-focused evaluation.

Core notation and bands. We use band operators (P_o, P_f, P_g, P_a) for clutter/DC, flow, guard, and alias regions, respectively. For short ensembles, these are exact orthogonal DFT-bin projectors; for longer apertures they are leakage-controlled analysis operators and are treated as band-energy statistics with explicit sanity sentinels (Supplementary Material).

2.1 Related context

Spatiotemporal SVD, RPCA, HOSVD, and local adaptive clutter filtering are established in ultrafast Doppler and fUS, and learned residualizers are relevant prior work on the residual-formation side of the pipeline [11, 12, 30, 31, 44]. The question here is narrower: once a residual slow-time cube has been fixed, can the downstream score head reduce nuisance leakage more effectively than PD- or Kasai-style readouts? That separates this paper from Yu-, Baranger-, and Song-style local clutter filtering, which adapts the clutter filter itself before a conventional Doppler score is applied [13, 14, 16]. ULM motion correction, twinkling-phantom structure audits, and hemodynamic fUS readouts therefore enter here as evaluation context and downstream audits rather than as the primary algorithmic contribution; the fuller literature context is kept in the methods companion.

2.2 Baseline clutter suppression and score conventions

Beamformed slow-time data. Let $X[t, y, x] \in \mathbb{C}$ denote a beamformed slow-time IQ movie over $t = 0, \ldots, T-1$ and spatial indices (y, x). We first apply a baseline clutter-suppression operator (e.g. motion-compensated SVD), producing a residual cube $X_{\text{base}}[t, y, x]$ used for both baseline scoring and as the input to the detector family.

Power Doppler proxy and baseline scores. From the baseline-filtered cube we define the baseline ROC score map

$$S_{\text{base}}(y, x) := \frac{1}{T} \sum_{t=0}^{T-1} |X_{\text{base}}(t, y, x)|^2 . \tag{1}$$

This is ordinary power Doppler on X_{base}. For short-ensemble regimes we additionally report auxiliary conventional comparators such as log-power Doppler and Kasai lag-1 magnitude [15] so that baseline-score idiosyncrasies do not dominate the comparison.
When the same power statistic is used only as a display or proxy map for deterministic preprocessing, we write

$$\text{PD}^{\uparrow}(y, x) \equiv S_{\text{base}}(y, x),$$

to make it explicit that the quantity is being used for visualization or mask construction rather than as the ROC score itself.

2.3 Score definitions

S_{base} denotes the baseline ROC score map on X_{base}, S_{det} the matched-subspace detector score map, and $\text{PD}^{\uparrow}$ a display or proxy map used only for visualization or deterministic preprocessing rather than for ROC scoring. Unless explicitly stated, the detector family consists of three detector variants built from the same band-limited score family: a fixed matched-subspace head without whitening, a prespecified adaptive variant that promotes selected pixels onto a whitened branch when guard-band evidence indicates clutter, and a fully whitened covariance-adaptive variant.

3 Method

What changes relative to SVD + PD. Relative to a conventional ultrafast Doppler pipeline, the proposed detector family changes the final scoring stage rather than replacing upstream clutter suppression. After forming a residual slow-time signal, we score its energy inside a prespecified flow subspace rather than using only aggregate Doppler power. The simplest fixed head uses that score without local whitening, a prespecified adaptive variant promotes clutter-heavy pixels onto a whitened branch only when guard-band evidence warrants it, and a fully whitened variant is retained for regimes where stronger local adaptation is beneficial. The goal is not simply to increase vessel conspicuity, but to reduce nuisance-driven high scores in operating regions where false positives matter.

Plain-language interpretation. The fixed head is a local band-limited flow detector applied after ordinary residualization. The adaptive branch adds tile-local whitening only where guard-band energy indicates localized clutter.

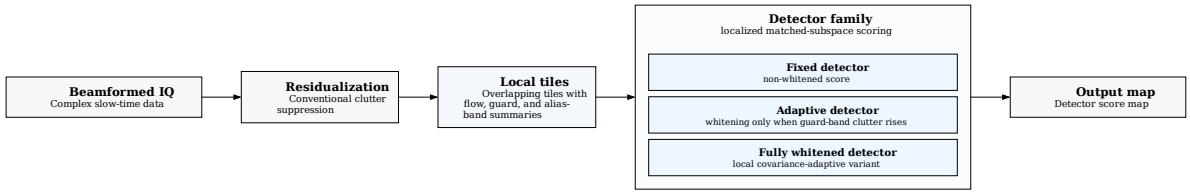

Figure 3: Method overview. Beamformed IQ first undergoes ordinary residualization, after which the detector operates tilewise in slow time. The fixed detector uses the band-limited matched-subspace score everywhere. The adaptive detector uses that same score by default and switches to a whitened branch only in tiles with strong guard-band evidence of localized clutter. The fully whitened detector applies whitening everywhere.

Pipeline overview.

1. Baseline clutter suppression (e.g. MC–SVD) produces a residual cube X_{base} and baseline score $S_{\text{base}}(y, x)$.
2. For each tile Ω_i, form the local slow-time support used by the chosen detector head; when whitening is active, apply the fixed regime-specific local covariance transform $y_{i,k} = R_i^{-1/2} x_{i,k}$.
3. Compute the self-normalized matched-subspace score on the chosen tile representation and aggregate over overlapping tiles to obtain the detector score map. The fixed and fully whitened forms are given explicitly in Eqs. (2) and (3); setting $R_i = I$ recovers the simpler band-limited fixed head.
4. If the adaptive variant is used, retain the fixed head by default and promote pixels onto the whitened branch only where a prespecified guard-band clutter feature exceeds a fixed threshold.

In practical terms, the method asks a local question after ordinary clutter suppression: within a small spatial neighborhood, does the remaining slow-time activity look more like microvascular flow or more like locally structured nuisance? Detector-core ablations on labeled brain simulations and the Gammex phantom are reported in the main Results.

The hierarchy is simple: fixed is the default method, adaptive selectively invokes whitening, and fully whitened applies whitening throughout. The derivations below formalize those three detector modes.

Three detector variants. On tile i, let $x_{i,k}$ denote the unwhitened slow-time snapshots and $y_{i,k} = R_i^{-1/2} x_{i,k}$ the whitened snapshots when whitening is active. The fixed detector is

$$S_0(i) := \frac{1}{K} \sum_{k=1}^{K} \frac{x_{i,k}^{\mathrm{H}} P_f \, x_{i,k}}{x_{i,k}^{\mathrm{H}} (I - P_f) \, x_{i,k} + \rho \, x_{i,k}^{\mathrm{H}} x_{i,k} + \epsilon}, \tag{2}$$

which is the same band-limited matched-subspace ratio with $R_i = I$. The fully whitened detector is

$$S_w(i) := \frac{1}{K} \sum_{k=1}^{K} \frac{y_{i,k}^{\mathrm{H}} P_f \, y_{i,k}}{y_{i,k}^{\mathrm{H}} (I - P_f) \, y_{i,k} + \rho \, y_{i,k}^{\mathrm{H}} y_{i,k} + \epsilon}. \tag{3}$$

The paper studies three concrete detector variants. The fixed detector uses $S_0(i)$ everywhere. The prespecified adaptive detector uses

$$S(i) = \begin{cases} S_w(i), & r_g^{\text{base}}(i) \geq \tau_g, \\ S_0(i), & r_g^{\text{base}}(i) < \tau_g, \end{cases} \tag{4}$$

where $r_g^{\text{base}}(i)$ is a baseline guard-band energy fraction and τ_g is fixed once and then held fixed across all reported experiments. The fully whitened detector uses $S_w(i)$ everywhere. In words, the adaptive detector keeps the simpler score in clean tiles and only turns on full whitening where the residual itself shows guard-band evidence of localized clutter.

Tile support and switching rule. All reported detectors operate on overlapping local tiles. In the main quantitative regimes, the spatial support is fixed at 8×8 pixels, with slow-time aperture $L_t = 8$ for kWave-Stress, $L_t = 16$ for Gammex-Struct, and $L_t = 64$ for SIMUS-Struct and ULM-CoreShell. The adaptive rule evaluates the baseline guard-band fraction $r_g^{\text{base}}(i)$ on each tile and switches from $S_0(i)$ to $S_w(i)$ only when $r_g^{\text{base}}(i) \geq \tau_g$; otherwise the fixed score is retained. Full details on sliding-window temporal embedding, covariance estimation, and conditioning are kept in the methods companion.

3.1 Algorithmic integrity and deterministic safeguards

To make the architecture auditable and to avoid leakage or discretization artifacts masquerading as performance gains, we enforce deterministic, label-free safeguards:

- **Conditional execution as a decision-rule variant.** To reduce latency, local whitening can be skipped outside a flow-proxy region and replaced by the baseline score there. Because this changes the overall decision rule, any conditional-execution reporting is treated as a distinct algorithmic variant and is accompanied by leakage controls (same-window vs. disjoint-window vs. randomized proxy masks). Unless otherwise stated, conditional execution uses *disjoint-window* proxy gating for detection reporting.
- **Short-ensemble band discretization sanity.** For extremely short ensembles (e.g. $N \in \{5, 9, 17\}$), the coarse frequency resolution $\Delta f = \text{PRF}/N$ can cause requested bands to collapse or overlap after discretization. Invalid discretizations are flagged explicitly and excluded from detector reporting.
- **Proxy-mask circularity control.** Here, proxy-mask circularity means reusing a proxy-derived mask both to steer conditional execution and to define the same quantitative endpoint being reported. Proxy maps are never used to compute ROC/TPR/FPR. Quantitative ROC claims are restricted to regimes with independent H_1/H_0 labels (simulator truth or B-mode structural geometry).

The methods companion records the corresponding reason labels and failure modes.

4 Experimental Design

4.1 Prespecified operating protocol and operating-point reporting

All processing settings (tile geometry, aperture, band definitions, and adaptive threshold) are fixed for each regime before held-out results are reported. Detection is evaluated by thresholding the

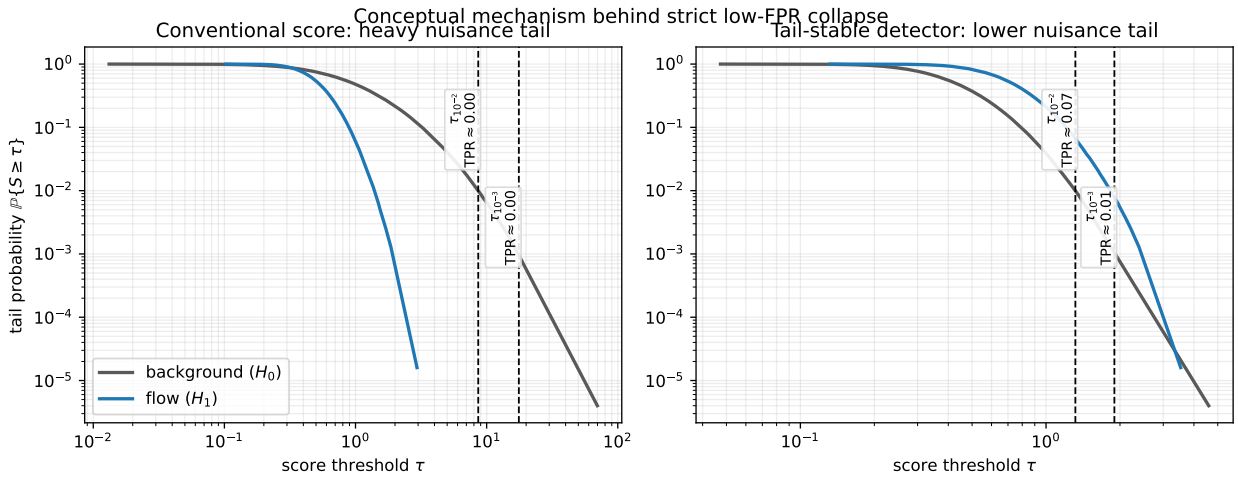

Figure 4: Conceptual strict-tail mechanism. Left: a conventional score can still show moderate-FPR separation while failing at a stricter operating point because the background right tail pushes the background-calibrated threshold $\tau_{10^{-3}}$ above essentially all flow pixels, yielding TPR ≈ 0 by construction. Right: when the nuisance tail is reduced, the same strict background allowance leaves some flow pixels above threshold. The later labeled-brain and Gammex tail figures are dataset-specific examples of this same mechanism.

high-score tail (right tail) at fixed FPR targets:

$$\mathbb{P}\{S \geq \tau_\alpha \mid H_0\} = \alpha, \qquad \mathrm{TPR}(\alpha) = \mathbb{P}\{S \geq \tau_\alpha \mid H_1\}.$$

For each experiment, we report the supported floor $fpr_{\min} = 1/n_{\mathrm{neg}}$ and bootstrap uncertainty at the natural unit of variation (windows or frames, depending on regime). These choices enforce a strict low-FPR operating discipline and avoid over-interpreting unsupported tail points. We emphasize $\alpha = 10^{-3}$–10^{-4} because those operating points expose nuisance-tail failure modes cleanly, not because routine fUS use must literally operate at those exact values. The same map-trustworthiness problem is already visible at a conventional matched background FPR of 10^{-2} in the worked Gammex example of Figure 2, where the baseline thresholded map is dominated by off-lumen detections while the matched-subspace score recovers lumen-consistent structure.

Threshold calibration and deployment. Unless otherwise stated, τ_α is calibrated per window (or per cine sequence) on that unit's negative set to report $\mathrm{TPR}(\alpha)$ at the intended operating point; in labeled regimes, the negative set is available for *exact* calibration, while in deployment it would be estimated via a negative-only training procedure or from a fixed calibration set. The Results section therefore includes two non-retrospective transfer checks: a fixed-calibration audit on SIMUS-Struct and a complementary block-split fixed-calibration audit on the ULM structural audit.

How the prespecified defaults were chosen. The numerical settings in this pipeline do not all play the same role, and they were not selected in the same way. Band edges are tied to acquisition physics: PRF fixes the discrete Doppler grid and Nyquist limit, while the flow/guard/alias partitions are chosen to separate slow parenchymal microvascular content from transition-band and aliased

high-velocity nuisance on that grid. Tile geometry and L_t are chosen once per regime to preserve spatial adaptation while retaining enough slow-time support for stable detector estimates, then held fixed. The adaptive threshold τ_g is a single frozen guard-energy cutoff chosen to preserve the fixed head in cleaner regimes while turning on whitening only when guard-band clutter evidence is clearly non-negligible. Full whitening-estimator and conditioning details are kept in the methods companion.

4.2 Integrity checks and deterministic sentinels

This work uses deterministic, label-free integrity checks for acquisition metadata, short-ensemble band discretization, proxy construction, and compute gating. Those checks either record provenance and proceed conservatively or exclude invalid configurations from reporting. The compact checklist and logged failure modes are kept in the methods companion.

4.3 Regimes and scope

The evidence is stratified by regime so that quantitative detection results remain clearly separated from robustness or transfer audits. The main scope is straightforward: downstream detector-head design matters after fixed residualization; the fixed band-limited head is the default structural choice on the held-out SIMUS benchmark; adaptive and fully whitened variants help in narrower clutter-heavy or structure-concentration regimes; and fixed-threshold transfer is encouraging in selected settings but not solved.

- **kWave-Stress.** These are controlled stress tests for strict-tail behavior. Across the open-skull, alias-dominant, and structured-clutter scenarios, the matched-subspace detector family retains measurable sensitivity at ultra-low FPR, with the strongest gains in the alias-dominant and structured-clutter settings.
- **Gammex-Struct.** These provide independent structure-based labels on real scanner data. We derive B-mode-only tube masks, compare conventional short-ensemble Doppler scores with matched-subspace detector-family scores, and report structural low-FPR operating points and within-ensemble motion ladders.
- **SIMUS-Struct.** This is the main held-out use-case-motivated synthetic benchmark in the paper. It uses the SIMUS/PyMUST forward model, but the evaluated settings are narrowed using real-data fUS use cases and observed operating characteristics rather than generic simulator coverage alone [8,9,34,45–47]. We report held-out structural and functional performance for the prespecified SIMUS-Struct-Mobile and SIMUS-Struct-Intraop regimes, together with a held-out clutter-heavy stress test on SIMUS-Struct-Mobile under cardiac-like pulsation and short ensembles.
- **ULM-CoreShell.** This dataset is used for motion-robustness analyses on real IQ, with controlled rigid and nonrigid motion injection and explicit fairness checks. We also report a structural audit using the published PALA super-resolution vascular rendering registered back onto the IQ grid, with a vessel-core versus perivascular-shell endpoint.
- **Shin-ProxyMotion.** This open real-IQ dataset provides a second label-free motion-stress evaluation under a prespecified low-frequency operating setting.
- **Supplementary analyses.** Additional label-free audits (e.g. PD-only whole-brain fUS) and diagnostic panels are provided in Supplementary Material.

Each regime uses one frozen operating profile for baseline, detector support, and Doppler-band geometry. The methods companion records the full regime-level whitening recipes and profile details.

Equation (1) defines the baseline ROC score, Equation (2) the fixed detector, Equation (4) the adaptive detector, and Equation (3) the fully whitened detector. Regime-specific whitening estimators, loading values, and training-support policies are fixed before reporting and are listed in the methods companion.

5 Results

Results roadmap. We first report the held-out SIMUS benchmark that determines the default detector configuration. We then show clutter-heavy stress tests that define when whitening helps. Next, we report mechanism ablations on labeled brain simulations and structurally labeled phantom data. Finally, we report bounded real-IQ structural and motion-robustness audits together with replay feasibility.

5.1 SIMUS-Struct benchmark

This moving-scatterer benchmark is intentionally narrower than a generic SIMUS parameter sweep. It retains the SIMUS/PyMUST moving-scatterer forward model [45, 46], but the evaluated settings are chosen to reflect whole-brain, cranial-window, mobile, and intra-operative fUS use cases rather than generic simulator realism alone [8, 9, 34, 47]. We evaluate two prespecified structural regimes, which we refer to jointly as SIMUS-Struct: SIMUS-Struct-Mobile and SIMUS-Struct-Intraop. Both were calibrated against real-data operating characteristics and then fixed before held-out testing. A third, more surface-dominated intra-operative regime remains exploratory and is not used for quantitative claims. This is a use-case-motivated synthetic benchmark with nuisance settings informed by reported fUS operating conditions. It does not substitute for prospective clinical validation. We report one frozen benchmark definition with explicit held-out splits and frozen positive, background, and nuisance masks. The structural checkpoint uses development seeds 125–126 to choose the final method families and settings, and held-out evaluation seeds 127–128 for the results in Table 4. The functional checkpoint uses a disjoint development split (seeds 221–222) and held-out evaluation split (seeds 223–224) under a common downstream readout. The evaluation script computes the reported nuisance-control metrics directly from those frozen masks under a fixed right-tail score convention.

For the structural task, the positive class is unaliased parenchymal Pf-band flow and the negative classes are (i) ordinary background and (ii) explicit nuisance structures such as superficial high-velocity or specular components. The three summary metrics used throughout this section are therefore plain-language clinical surrogates: $AUC_{\text{main/bg}}$ measures separation of target flow from ordinary background, $AUC_{\text{main/nuis}}$ measures separation of target flow from clinically relevant nuisance, and FPR_{nuis} @ $TPR_{\text{main}} = 0.5$ asks how many nuisance detections survive when the detector is tuned to retain half of the target positives. This last quantity is the most direct nuisance-tail metric in the table.

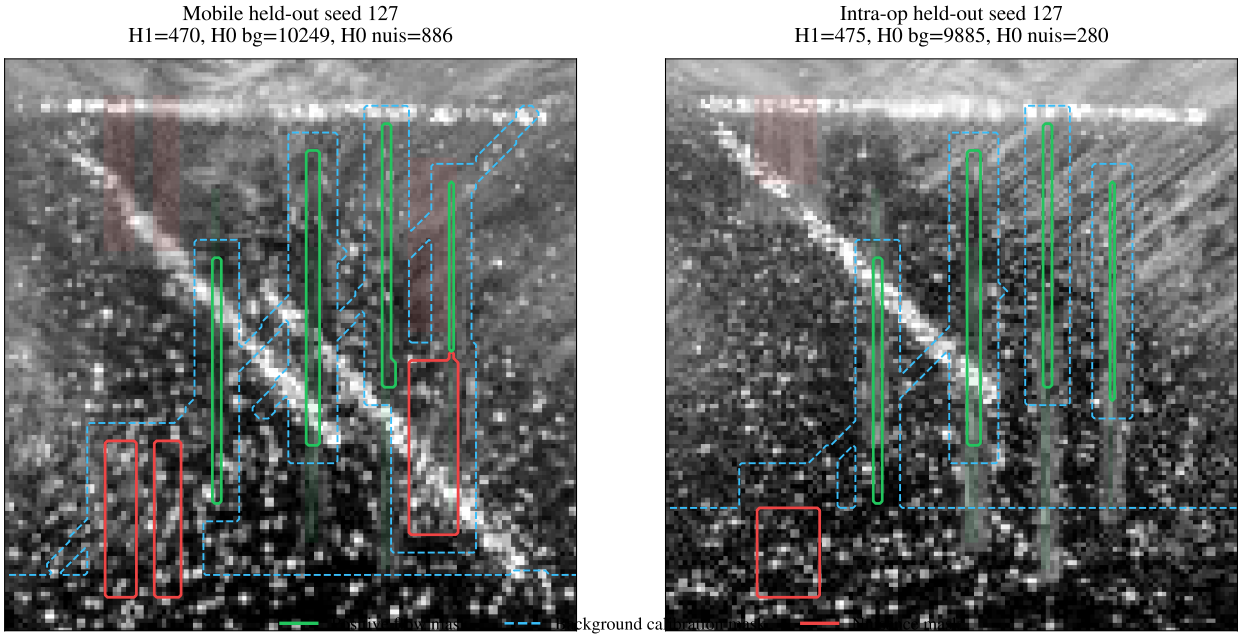

Figure 5: Representative SIMUS-Struct geometry for the SIMUS-Struct-Mobile and SIMUS-Struct-Intraop reference regimes. The grayscale underlay is log beamformed power from the simulated IQ cube and is shown only to make the mask geometry visible. Green contours mark the parenchymal-flow positive set H_1, blue dashed contours mark the broad background pool H_0^{bg} used to set the score threshold, and red contours mark the nuisance-negative set H_0^{nuis} used to quantify structured false detections at that same operating point. Table 4 reports the held-out seed results on these frozen mask types across seeds 127–128.

Figure 5 makes the benchmark contract concrete. The threshold is set on a broad off-flow background pool, while the nuisance column asks whether any detections survive in a smaller artifact-prone region at that same operating point. The later stress rows keep this same basic contract but make the nuisance environment harder by adding stronger pulsation and shorter support.

For the functional task, we use the same prespecified settings and a common background-controlled GLM-style readout on simulated task/null runs. We refer to that experiment as the SIMUS functional audit. Two comparisons are reported because they answer different questions: end-to-end pipeline ranking asks which full processing chain performs best after all residualizer-setting interactions, whereas the same-cube downstream-head comparison asks whether the matched-subspace detector improves the *same* residual cube under the same downstream readout.

We begin with the held-out reference regime that defines the default downstream head.

Mobile

Best conventional baseline: HOSVD → Kasai. Fixed detector chain: Adaptive Global SVD → fixed matched-subspace.

Seed	Best conventional baseline			Fixed detector chain		
	$\mathrm{AUC}_{\mathrm{main/bg}}$	$\mathrm{AUC}_{\mathrm{main/nuis}}$	$\mathrm{FPR}_{\mathrm{nuis}}$ @ $\mathrm{TPR}_{\mathrm{main}} = 0.5$	$\mathrm{AUC}_{\mathrm{main/bg}}$	$\mathrm{AUC}_{\mathrm{main/nuis}}$	$\mathrm{FPR}_{\mathrm{nuis}}$ @ $\mathrm{TPR}_{\mathrm{main}} = 0.5$
Seed 127	0.647	0.624	0.361	0.798	0.923	0.000
Seed 128	0.692	0.720	0.210	0.905	0.984	0.000

Intra-operative parenchymal

Best conventional baseline: HOSVD → Kasai. Fixed detector chain: RPCA → fixed matched-subspace.

Seed	Best conventional baseline			Fixed detector chain		
	$\mathrm{AUC}_{\mathrm{main/bg}}$	$\mathrm{AUC}_{\mathrm{main/nuis}}$	$\mathrm{FPR}_{\mathrm{nuis}}$ @ $\mathrm{TPR}_{\mathrm{main}} = 0.5$	$\mathrm{AUC}_{\mathrm{main/bg}}$	$\mathrm{AUC}_{\mathrm{main/nuis}}$	$\mathrm{FPR}_{\mathrm{nuis}}$ @ $\mathrm{TPR}_{\mathrm{main}} = 0.5$
Seed 127	0.834	0.431	0.643	0.993	0.99998	0.000
Seed 128	0.839	0.471	0.600	0.995	0.99994	0.000

Table 4: Held-out SIMUS-Struct reference regimes with explicit held-out seed rows. For each evaluated structural setting, we report the best conventional baseline from the development-split baseline sweep and the strongest fixed detector chain chosen on development seeds 125–126, then evaluated unchanged on held-out seeds 127–128. Seed-specific held-out values are shown directly rather than pooled summaries. Here, 0.000 means exactly zero observed nuisance detections on the frozen held-out nuisance masks for that seed, not a rounded small value; the supported floor is set by the corresponding nuisance-mask size.

Table 4 is the held-out reference regime. On the mobile setting, the best conventional baseline among evaluated methods is HOSVD → Kasai, while the strongest fixed detector chain chosen on development seeds 125–126 and evaluated unchanged on held-out seeds 127–128 is Adaptive Global SVD → fixed matched-subspace. On the intra-operative parenchymal setting, the best conventional baseline is again HOSVD → Kasai, while the corresponding fixed detector chain is RPCA → fixed matched-subspace. Figure 5 shows the mask geometry used for these audits: thresholds are set on a broad background pool and nuisance leakage is measured on compact structured artifact zones. The fixed-chain rows suppress those nuisance zones completely on all four held-out seed-setting pairs, and the table reports the seed-specific held-out values directly. In Table 4, the 0.000 nuisance rows are literal zero observed nuisance detections on the held-out masks, not rounded near-zero rates. This is the main benchmark result: the band-limited fixed matched-subspace head is the strongest fixed default on held-out structural data, while Table 7 shows where the adaptive and fully whitened variants help in clutter-heavier regimes.

To isolate that detector-head claim directly on the main benchmark, Table 5 fixes one held-out MC–SVD residual stream and swaps only the downstream score head. This removes residualizer choice from the comparison. On that fixed residual, PD and Kasai still rank nuisance highly, the fixed matched-subspace head is the strongest same-residual detector, and the fully whitened detector is worse on both held-out settings. The adaptive detector is numerically identical to the fixed head here because its frozen guard-promotion rule does not activate on these cleaner MC–SVD residuals. On this benchmark, the adaptive extension therefore preserves the fixed default when the clutter evidence is weak instead of forcing whitening where it does not help.

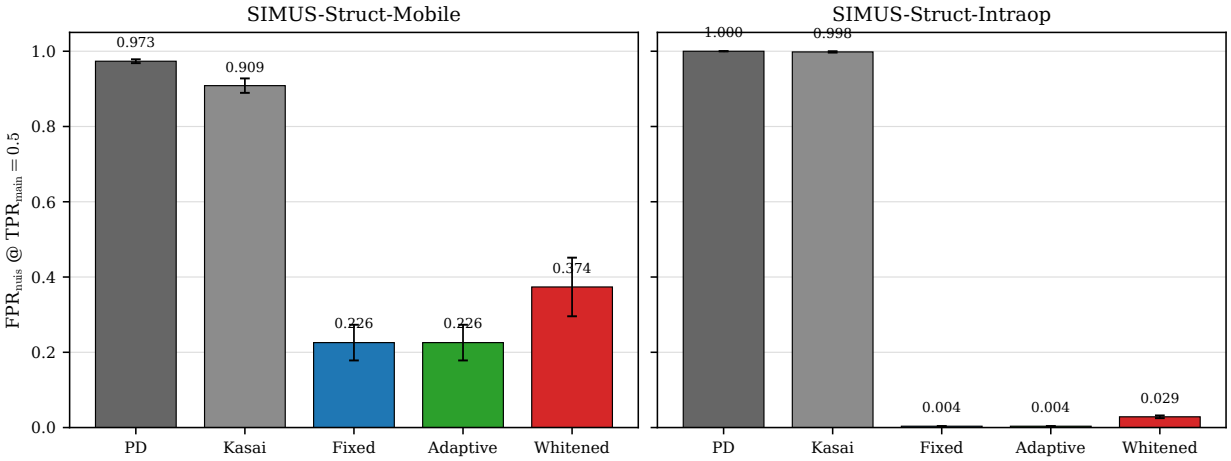

Figure 6: Headline same-residual downstream-head comparison on the held-out SIMUS reference benchmark using the common MC–SVD residual from Table 5. Bars show nuisance false-positive rate at matched main-response recall 0.5, the clearest operating-point summary of the benchmark. Lower bars are better, and the fixed matched-subspace head, matched by the adaptive variant under this frozen guard rule, is the strongest same-residual downstream choice in both settings. $\mathrm{AUC}_{\mathrm{main/nuis}}$ values are reported in Table 5.

Method	Mobile			Intra-operative parenchymal		
	$AUC_{main/bg}$	$AUC_{main/nuis}$	FPR_{nuis} @ $TPR_{main} = 0.5$	$AUC_{main/bg}$	$AUC_{main/nuis}$	FPR_{nuis} @ $TPR_{main} = 0.5$
Baseline (power Doppler)	0.568 [0.562,0.575]	0.131 [0.127,0.134]	0.973 [0.968,0.979]	0.912 [0.909,0.915]	0.022 [0.016,0.028]	1.000 [1.000,1.000]
Baseline (Kasai lag-1 magnitude)	0.500 [0.489,0.512]	0.171 [0.154,0.188]	0.909 [0.889,0.928]	0.792 [0.788,0.795]	0.019 [0.017,0.022]	0.998 [0.996,1.000]
Fixed matched-subspace detector	0.623 [0.587,0.659]	0.713 [0.677,0.748]	0.226 [0.178,0.273]	0.844 [0.831,0.857]	0.982 [0.981,0.983]	0.004 [0.004,0.004]
Adaptive detector	0.623 [0.587,0.659]	0.713 [0.677,0.748]	0.226 [0.178,0.273]	0.844 [0.831,0.857]	0.982 [0.981,0.983]	0.004 [0.004,0.004]
Fully whitened detector	0.569 [0.553,0.585]	0.600 [0.529,0.672]	0.374 [0.296,0.451]	0.819 [0.819,0.820]	0.932 [0.927,0.936]	0.029 [0.025,0.032]

Table 5: Same-residual detector-family ablation on SIMUS-Struct using a common MC–SVD residual. Rows differ only in the downstream score head: power Doppler, Kasai lag-1 magnitude, the fixed flow-band matched-subspace detector without whitening ($R = I$), the adaptive detector that promotes clutter-heavy tiles onto a whitened branch, and the fully whitened detector. Values are means with [min,max] over held-out evaluation seeds 127 and 128 for each setting. Lower nuisance FPR at matched TPR is better.

The key practical question is not whether the method makes vessels look better on average, but whether fewer nuisance detections survive when the detector is tuned to preserve clinically relevant parenchymal flow. That is the practical meaning of FPR_{nuis} @ $TPR_{\text{main}} = 0.5$: it asks how much structured false signal remains at a matched target-flow sensitivity operating point. Because fUS is already being used in intra-operative, bedside, cranial-window, and emerging mobile settings [1, 8, 20–22], that nuisance-leakage tradeoff is tied directly to usability and reader trust rather than to image aesthetics alone.

That per-window operating-point protocol is still an evaluation protocol, not a scanner-time thresholding rule. To address the more practical question directly, we also ran a fixed-calibration transfer check in which the threshold is learned once from separate calibration seeds 125–126 and then applied unchanged to the held-out evaluation seeds 127–128. Table 6 reports that result at nominal $\alpha = 10^{-3}$. On SIMUS-Struct-Mobile, the fixed matched-subspace head preserves nonzero held-out main recall (0.004 [0.002, 0.006]) while lowering realized nuisance FPR from 0.016 [0.015, 0.018] for PD to 0.003 [0.000, 0.006]. On SIMUS-Struct-Intraop, both methods remain near the nominal background target, but only the fixed head suppresses nuisance leakage to 0.000 [0.000, 0.000], whereas PD remains at 0.489 [0.464, 0.514]. This is not the full prospective scanner workflow yet, but it does show that the fixed default from the held-out reference regime survives a separate-bank threshold transfer on the benchmark that anchors the paper.

Method	Mobile			Intra-operative parenchymal		
	TPR_{main}	realized FPR_{bg}	realized FPR_{nuis}	TPR_{main}	realized FPR_{bg}	realized FPR_{nuis}
Power Doppler on MC–SVD residual	0.000 [0.000,0.000]	0.002 [0.000,0.003]	0.016 [0.015,0.018]	0.012 [0.006,0.017]	0.001 [0.001,0.001]	0.489 [0.464,0.514]
Fixed matched-subspace head	0.004 [0.002,0.006]	0.002 [0.001,0.003]	0.003 [0.000,0.006]	0.006 [0.004,0.008]	0.001 [0.001,0.001]	0.000 [0.000,0.000]

Table 6: Fixed-calibration transfer on SIMUS-Struct at nominal $\alpha = 10^{-3}$. For each setting, a single threshold is learned once from pooled background negatives on separate calibration seeds 125–126 and then applied unchanged to the held-out evaluation seeds 127–128. Entries report mean [min,max] across the two held-out seeds, so this table is a direct non-retrospective threshold-transfer check rather than the per-window oracle calibration used for ROC reporting elsewhere.

We then turn to harsher nuisance settings on the same benchmark family. These held-out stress rows use the same benchmark infrastructure but push it toward two clinically grounded nuisance regimes: cardiac-like low-frequency tissue pulsation and progressively shorter slow-time support. The pulsation rows use a fixed 1.30 Hz cardiac-like nuisance term whose amplitude is

increased over reference, moderate, and hard settings; this probes the same low-frequency tissue-motion regime emphasized by Demené *et al.* as critical for SVD clutter filtering and by Baranger *et al.* in neonatal/open-skull Doppler settings [11,12]. The short-ensemble rows reduce the support from 64 frames to 48 and 32 frames to mimic clinically constrained acquisitions in which shorter windows are used to limit motion corruption or workflow burden [1,22,34]. This analysis focuses on the residualizer families that dominated the preliminary reduced-grid search (RPCA and adaptive-global SVD) and then reruns the held-out evaluation at the full reporting tier.

Cardiac-like pulsation

Level / perturbation	Baseline (RPCA→PD)	Fixed head	Adaptive head	Whitened head
Ref.	$0.381^{0.416}_{0.345}$	$0.879^{0.898}_{0.860}$	$0.879^{0.898}_{0.860}$	$0.911^{0.925}_{0.894}$
1.0x cardiac amplitude at 1.30 Hz	$0.743^{0.842}_{0.652}$	$0.034^{0.047}_{0.023}$	$0.034^{0.046}_{0.023}$	$0.017^{0.027}_{0.008}$
Mod.	$0.388^{0.423}_{0.354}$	$0.877^{0.895}_{0.857}$	$0.877^{0.895}_{0.857}$	$0.903^{0.920}_{0.886}$
2.0x cardiac amplitude at 1.30 Hz	$0.668^{0.795}_{0.593}$	$0.049^{0.064}_{0.030}$	$0.049^{0.064}_{0.030}$	$0.030^{0.043}_{0.019}$
Hard	$0.399^{0.433}_{0.365}$	$0.875^{0.893}_{0.854}$	$0.875^{0.894}_{0.854}$	$0.905^{0.921}_{0.890}$
3.0x cardiac amplitude at 1.30 Hz	$0.609^{0.770}_{0.537}$	$0.044^{0.058}_{0.029}$	$0.044^{0.059}_{0.029}$	$0.027^{0.041}_{0.017}$

Short ensemble

Level / perturbation	Baseline (RPCA→PD)	Fixed head	Adaptive head	Whitened head
Ref.	$0.381^{0.415}_{0.345}$	$0.879^{0.897}_{0.861}$	$0.879^{0.898}_{0.859}$	$0.911^{0.926}_{0.895}$
64 frames (42.7 ms)	$0.743^{0.840}_{0.649}$	$0.034^{0.046}_{0.023}$	$0.034^{0.047}_{0.023}$	$0.017^{0.027}_{0.008}$
Mod.	$0.390^{0.423}_{0.356}$	$0.784^{0.807}_{0.758}$	$0.784^{0.808}_{0.759}$	$0.846^{0.868}_{0.824}$
48 frames (32.0 ms)	$0.754^{0.814}_{0.708}$	$0.137^{0.169}_{0.102}$	$0.137^{0.167}_{0.103}$	$0.058^{0.080}_{0.034}$
Hard	$0.426^{0.462}_{0.392}$	$0.772^{0.796}_{0.746}$	$0.772^{0.797}_{0.746}$	$0.716^{0.745}_{0.686}$
32 frames (21.3 ms)	$0.617^{0.727}_{0.536}$	$0.166^{0.194}_{0.139}$	$0.166^{0.193}_{0.139}$	$0.160^{0.207}_{0.130}$

Table 7: Concrete detector-head view of the held-out SIMUS mobile stress regime, fixing the residualizer to RPCA on every row. Entries report $\mathrm{AUC}_{\mathrm{main/nuis}}$ (top) and $\mathrm{FPR}_{\mathrm{nuis}}$@0.5 (bottom), with point estimates shown as $x^{\mathrm{hi}}_{\mathrm{lo}}$ using 95% nonparametric bootstrap intervals over the held-out masked score samples. RPCA is used because RPCA → PD is the strongest conventional baseline among evaluated methods on every reported stress row. This table separates the fixed, adaptive, and fully whitened heads directly instead of reporting only the best family member on each row.

Table 7 is the main held-out stress result. It fixes the residualizer to RPCA and reports the fixed, adaptive, and fully whitened heads separately on the same held-out stress rows. The pattern is coherent without any row-wise family summary: the fully whitened head is strongest on the cardiac-pulsation rows and on the 64-frame and 48-frame short-ensemble rows, while the fixed head remains strongest only on the hardest 32-frame row. On these reduced stress rows the adaptive head is numerically identical to the fixed head, so the family-level advantage is not coming from freely switching among all three modes on every row. A row-wise best-family summary against the strongest conventional baseline is provided in the Supplementary Material. Because this stress regime uses one held-out evaluation seed per axis, the uncertainty notation here is a within-seed bootstrap interval over the held-out masked scores rather than a between-seed interval.

The stress rows complement the reference regime. Together, the stress rows show nontrivial difficulty and the reference rows identify which fixed head should be deployed by default.

Functional same-cube detector-head comparison. The structural nuisance-control results remain primary, but the paper also reports one held-out functional same-cube audit because it

answers the same downstream-head question under a common activation-versus-nuisance readout. Table 8 fixes the residual stream and compares the detector heads under the same functional readout. The fully whitened detector is directionally positive across most residual families in both settings, but the effect size is smaller and less decisive than in the structural benchmark, so we treat it as secondary supporting evidence rather than as a headline result.

Benchmark setting	Best end-to-end chain	Matched-subspace-positive families	Median Δ selection score [IQR]	Median $\Delta\mathrm{AUC}_{\mathrm{act/nuis}}$ (IQR)
Intra-operative parenchymal	RPCA $\rightarrow$ PD after matched-subspace	5/6	+0.065 [+0.032,+0.091]	+0.026 [+0.014,+0.052]
Mobile	HOSVD $\rightarrow$ PD	5/6	+0.111 [+0.032,+0.142]	+0.059 [+0.011,+0.066]

Table 8: Secondary held-out functional summary on the prespecified SIMUS-Struct settings. This table uses one common background-controlled activation readout for every compared pipeline, so the rows differ only in the upstream residualizer and downstream detector head. "Best end-to-end chain" is the strongest full held-out pipeline on that setting under that common readout. "Matched-subspace-positive families" counts how many of the six prespecified residual families improved when the matched-subspace detector stage replaced the native conventional score on the same residual cube. "PD after matched-subspace" denotes a power-Doppler-style summary map formed after the matched-subspace detector stage. Δ selection score is the held-out pipeline-ranking metric used in this functional benchmark, reported as matched-subspace minus native. $\Delta\mathrm{AUC}_{\mathrm{act/nuis}}$ is the corresponding change in separation between simulated activation and nuisance regions under the same readout. Both deltas are summarized by the median and interquartile range across the six residual families.

Scope and low-FPR operating protocol. Quantitative detection performance is reported on kWave-Stress, Gammex-Struct, and SIMUS-Struct using the strict low-FPR operating protocol of Section 4.1. The ULM and Shin sections add structural, robustness, and transfer audits on open real-IQ data. Conditional execution is a decision-rule variant that uses a flow-proxy mask; leakage ablations and additional diagnostics are reported in Supplementary Material.

5.2 Brain simulation stress tests

The SIMUS-Struct benchmark above is the held-out benchmark that anchors the paper. This subsection uses simpler labeled brain simulations to isolate detector behavior under controlled nuisance stress tests. We report low-FPR operating points on these scenarios using flow-evidence scores under a right-tail convention (higher = more flow-like; thresholds calibrated on the high-score tail): a baseline power Doppler score and the matched-subspace score (Notation). Baseline methods compare clutter suppression strategies (MC–SVD; adaptive SVD with a spatial singular-vector similarity cutoff; block-wise local SVD with overlap-add; RPCA; HOSVD) and are scored using the same power Doppler statistic; the matched-subspace detector family is evaluated on top of the MC–SVD residual. Across these scenarios, the PD-style baselines degrade sharply in the ultra-low-FPR tail, while the localized band-limited matched-subspace score retains measurable sensitivity at $\alpha = 10^{-4}$–10^{-3}, including in the structured-clutter setting where nuisance deliberately leaks

into the positive class. We keep the two clinically interpretable simulation settings here: an open-skull surface-artifact stress test and a structured-clutter leakage stress test. A third alias-dominant positive control showed the same qualitative trend and is reported in the Supplementary Material. Tables 9–11 report component audits on these labeled stress tests. Importantly, the MC–SVD+PD baseline row and the matched-subspace row share the identical MC–SVD residual cube and differ only in the downstream score (PD versus matched-subspace), isolating the effect of scoring rather than the clutter-suppression stage. This is why the detector-component and detector-swap tables belong here: they explain *why* the main SIMUS-Struct benchmark later favors certain processing chains, but they do not replace that benchmark. A supplementary threshold-transfer audit on the labeled brain simulation stress tests still shows that the robustly whitened variant remains close to target in both the open-skull and structured-clutter settings. Those simulation-only checks are secondary to the stronger deployment-transfer results on SIMUS-Struct and on ULM-CoreShell.

Explicit limitations from the labeled-brain audits. These labeled-brain audits do not make full whitening the structural default. On the prespecified held-out structural benchmark, the simpler band-limited matched-subspace detector remains stronger on nuisance-control metrics. On the labeled-brain stress tests, however, the fully whitened variant remains stable: open-skull realized FPR stays near target under both pairwise cross-window transfer (median 1.10×10^{-4}, $q_{75} = 1.64 \times 10^{-4}$ at $\alpha = 10^{-4}$) and a leave-one-window-out fixed calibration bank (median 1.20×10^{-4}), and the operating-point results are numerically unchanged across covariance-training trim $q \in \{0.05, 0.08, 0.10\}$. The open question is therefore where whitening earns its extra complexity, not whether the fully whitened variant is broadly unstable.

Baseline fairness and prespecified settings. All baseline hyperparameters are fixed prior to reporting results. For the labeled brain simulation scenarios, the MC–SVD energy-fraction cutoff is selected once and then held at $e = 0.90$ for all reported windows. The comparator families span global low-rank filtering (MC–SVD and adaptive-global SVD), block-wise local filtering (fixed-energy and adaptive-local SVD), RPCA, and HOSVD. Baseline rows are then evaluated with the same right-tail Doppler score so that differences reflect the residualizer rather than a change in the downstream scoring rule. We do *not* retune these baseline families on the held-out windows to maximize their own strict-tail TPR after seeing the test cases; the main SIMUS-Struct benchmark instead gives each family a development-split configuration-selection step before held-out reporting, while the labeled brain stress tests use fixed representative settings chosen once and then frozen. The compact family summary is provided in Supplementary Material; here the emphasis is on what the comparisons mean scientifically rather than on implementation taxonomy.

Detector-component ablation (whitening versus band selectivity). To address the concern that the proposed score is merely "local whitening plus a ratio" under a new name, we make the mechanism check explicit. On the identical MC–SVD residual cube, we compare the proposed score to two deliberately simple downstream comparators: (i) *whitened total power* (apply $R_i^{-1/2}$ but do not partition the Doppler spectrum) and (ii) an *unwhitened* matched-subspace ratio (set $R_i = I$ but retain the fixed flow-band subspace). Table 9 is the key mechanism check. Whitening alone without band partition stays at zero across the reported strict-tail operating points, while the unwhitened band-limited ratio remains strong in both labeled stress tests. In the labeled-brain stress tests, the gain therefore appears to come from matched-subspace geometry rather than from whitening alone.

We interpret the contribution accordingly: the novelty here is downstream detector-head design and evaluation discipline, with local covariance adaptation as one useful extension rather than the whole story.

Method	Open-skull surface-artifact stress test			Structured-clutter leakage stress test		
	$\alpha = 10^{-4}$	$\alpha = 3\times10^{-4}$	$\alpha = 10^{-3}$	$\alpha = 10^{-4}$	$\alpha = 3\times10^{-4}$	$\alpha = 10^{-3}$
PD on MC–SVD residual	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
Whitened power (no band)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
Unwhitened ratio (no whitening)	0.8689 (0.8674,0.8689)	0.8947 (0.8831,0.8985)	0.9389 (0.9166,0.9400)	0.9104 (0.8900,0.9189)	0.9373 (0.9058,0.9389)	0.9619 (0.9512,0.9689)
Fully whitened matched-subspace detector	0.0730 (0.0008,0.0865)	0.1288 (0.0012,0.2222)	0.2218 (0.0015,0.4368)	0.5871 (0.5006,0.6682)	0.7205 (0.6048,0.7785)	0.8731 (0.8189,0.8758)

Table 9: Detector ablations on labeled brain simulation stress tests (MC–SVD residual, prespecified settings). All rows share the identical MC–SVD residual cube and differ only in the detector statistic: baseline PD, total power after tile-local whitening (no Doppler band partition), matched-subspace ratio without whitening (R=I), and the fully whitened matched-subspace detector. The two column groups correspond to an open-skull surface-artifact stress test and a structured-clutter leakage stress test. Numbers are medians (IQR) over five disjoint 64-frame windows (offsets 0/64/128/192/256).

Detector-swap composability audit (matched-subspace scoring on competing residuals). To isolate detector effects from the clutter-suppression stage, we also compute the matched-subspace detector score on residuals produced by competing clutter filters (RPCA and HOSVD) and compare it to power Doppler scoring on the same residual under the identical window and mask protocol. Table 10 reports this paired-residual audit. The labeled simulation detector-swap audit isolates detector-head behavior on controlled k-Wave stress tests under fixed residual streams, whereas Table 4 reports end-to-end performance on the held-out structural benchmark. Under the prespecified settings used throughout, strict-tail improvements do not automatically persist when the matched-subspace detector is composed with more aggressive low-rank baselines. In the main brain-simulation pipeline, the matched-subspace detector is therefore used as a second-stage detector on a coarse MC–SVD residual, and the controlled detector swap on that same residual provides the cleanest isolation of detector effects.

Method	Open-skull surface-artifact stress test			Structured-clutter leakage stress test		
	$\alpha = 10^{-4}$	$\alpha = 3\times10^{-4}$	$\alpha = 10^{-3}$	$\alpha = 10^{-4}$	$\alpha = 3\times10^{-4}$	$\alpha = 10^{-3}$
PD on RPCA residual	0.0000 (0.0000,0.0046)	0.0015 (0.0000,0.0088)	0.0042 (0.0042,0.0173)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
Matched-subspace detector on RPCA residual	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
PD on HOSVD residual	0.0000 (0.0000,0.0015)	0.0015 (0.0015,0.0015)	0.0046 (0.0046,0.0077)	0.0000 (0.0000,0.0015)	0.0000 (0.0000,0.0015)	0.0015 (0.0000,0.0031)
Matched-subspace detector on HOSVD residual	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0004)	0.0019 (0.0004,0.0019)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)

Table 10: Detector-swap fairness check on labeled brain simulation stress tests: PD vs matched-subspace detector scores computed on the *same* residual produced by a competing clutter filter (paired bundles). The two column groups correspond to an open-skull surface-artifact stress test and a structured-clutter leakage stress test. Numbers are medians (IQR) over five disjoint 64-frame windows (offsets 0/64/128/192/256).

Robust-whitening stability audit (labeled brain simulations). Because the fully whitened variant relies on tile-local covariance estimation, Table 11 tests sensitivity to contaminated training support and threshold calibration by varying only the high-energy covariance-training trim over $q \in \{0.05, 0.08, 0.10\}$. The labeled-brain operating-point results are numerically unchanged across those trim settings, and open-skull realized FPR remains close to target under both pairwise cross-window transfer and a fixed calibration bank. In these stress tests, contamination-robust covariance

training behaves as a stable recipe choice rather than as a dominant source of result variance. We further report a 1D sensitivity sweep over covariance diagonal loading λ, tile geometry, and slow-time aperture L_t on the canonical open-skull pilot (Supplementary Material).

Trim setting	Open-skull stress test			Structured-clutter stress test		
	Within TPR@10^{-4}	Cross FPR med. (q75)	Fixed-cal FPR med. (q75)	Within TPR@10^{-4}	Cross FPR med. (q75)	Fixed-cal FPR med. (q75)
$q = 0.05$	0.852	1.10e-04 (1.64e-04)	1.20e-04 (1.39e-04)	0.917	9.96e-05 (2.44e-04)	7.97e-05 (1.99e-04)
$q = 0.08$	0.852	1.10e-04 (1.64e-04)	1.20e-04 (1.39e-04)	0.917	9.96e-05 (2.44e-04)	7.97e-05 (1.99e-04)
$q = 0.10$	0.852	1.10e-04 (1.64e-04)	1.20e-04 (1.39e-04)	0.917	9.96e-05 (2.44e-04)	7.97e-05 (1.99e-04)

Table 11: Robust-whitening stability audit for the prespecified fully whitened detector on the labeled brain stress tests. The only changing parameter is the high-energy covariance-training trim q. Entries report within-window TPR at the strictest operating point ($\alpha = 10^{-4}$) together with realized FPR under pairwise cross-window transfer and a leave-one-window-out fixed calibration bank. The reported values are numerically unchanged across $q \in \{0.05, 0.08, 0.10\}$, indicating that the finalized advanced recipe is stable to modest covariance-training trim variation in these stress tests.

Relaxed operating-point sanity anchor (labeled brain simulations). Several baseline rows in the supplementary labeled-brain strict-tail table take the value $\mathrm{TPR}(\alpha) \approx 0$ at $\alpha \leq 10^{-3}$. To keep the main Results focused on SIMUS-Struct and the strongest labeled stress tests, we place the relaxed-operating-point sanity table ($\alpha \in \{10^{-3}, 10^{-2}, 10^{-1}\}$) in the Supplementary Material and retain here only the representative tail-collapse visualization that explains why those strict-tail zeros occur.

Why PD-after-filter baselines can collapse in the strict tail. Figure 4 shows the general mechanism: a baseline clutter filter can still show moderate-FPR separation yet fail at ultra-low FPR if the background score distribution has an extreme right tail. Then the background-calibrated threshold τ_α can lie above essentially all H_1 pixels, yielding $\mathrm{TPR}(\alpha) \approx 0$ by construction. In a representative open-skull labeled brain window the MC–SVD+PD baseline background quantiles are $q_{0.99} \approx 5.9$ and $q_{0.999} \approx 9.3$ while the flow $q_{0.999} \approx 0.094$, so no flow pixels exceed the $\alpha = 10^{-3}$ threshold. Figure 7 is one dataset-specific example of that geometry on the labeled brain benchmark. This tail-dominance failure mode motivates both the strict low-FPR operating discipline and detector heads that explicitly reshape the tail, beginning with band-limited matched-subspace scoring and, when needed, tile-local covariance whitening.

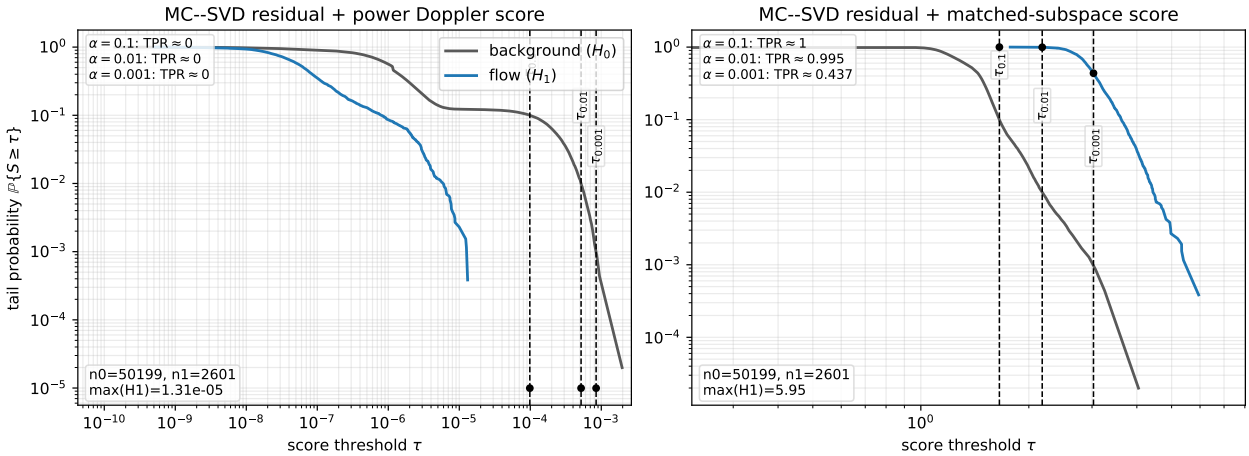

Figure 7: Representative open-skull labeled brain window illustrating strict-tail collapse of PD-after-filter baselines. Curves show empirical tail probabilities $\mathbb{P}\{S \geq \tau\}$ for background (H_0) and flow (H_1) score distributions versus score threshold τ, for (left) the baseline power Doppler score on the MC–SVD residual and (right) the matched-subspace score computed on the same residual. Vertical lines mark background-calibrated thresholds τ_α at $\alpha \in \{10^{-1}, 10^{-2}, 10^{-3}\}$ and markers show the corresponding $\mathrm{TPR}(\alpha) = \mathbb{P}\{S \geq \tau_\alpha \mid H_1\}$. For the baseline PD score, $\tau_{10^{-3}}$ lies above $\max(H_1)$, yielding TPR ≈ 0 by construction; the matched-subspace score retains nonzero sensitivity at the same strict-tail operating point.

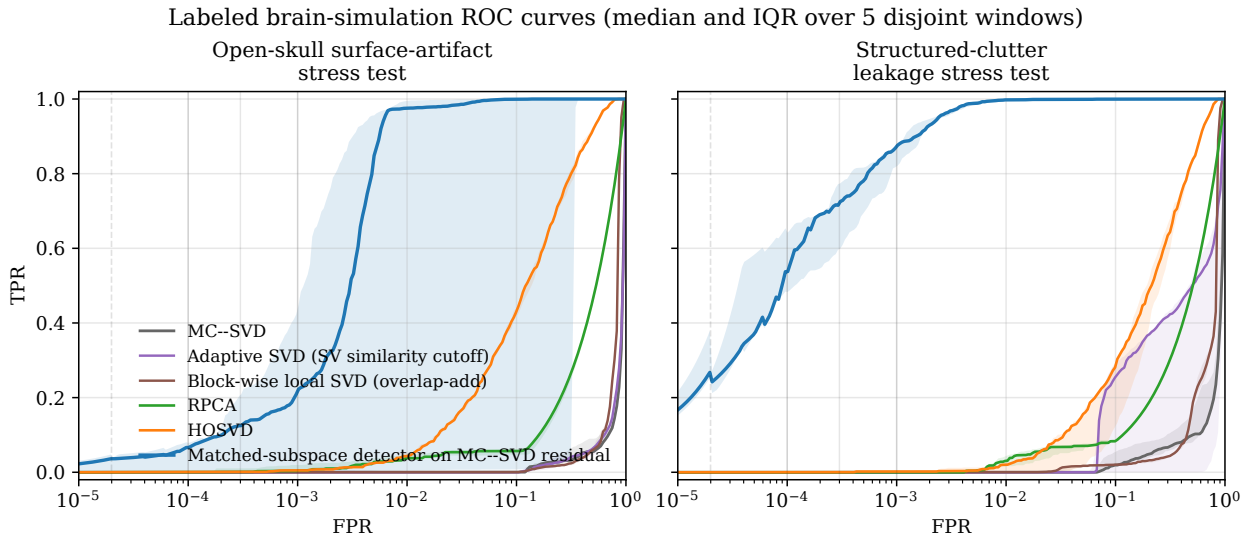

Figure 8: Controlled labeled-brain stress tests: ROC-style curves (TPR vs FPR on a log-x axis) under the same 64-frame window protocol used for the low-FPR operating-point reporting above. Curves include MC–SVD, adaptive and local SVD baselines, RPCA, HOSVD, and the matched-subspace detector; each curve shows the median and interquartile range over five disjoint windows. For readability, this figure keeps the two more clinically interpretable simulation settings (open-skull surface artifact and structured-clutter leakage); the alias-dominant positive control is reported in the Supplementary Material. The moderate-FPR region illustrates that several PD-after-filter baselines can show separation at relaxed operating points while saturating at strict-tail operating points. The vertical dashed line indicates the supported floor $fpr_{\min} = 1/n_{\text{neg}}$.

5.3 Gammex phantom: structural ROC and within-ensemble motion ladders

Dataset and acquisition parameters. We use the open-access Doppler Twinkling Artifact database [42], which provides complex beamformer output from phantom acquisitions on a clinical scanner. The *Gammex flow phantom* provides a simple vessel-like geometry that lets us derive independent structure-based flow/background labels from B-mode anatomy alone. Each cine frame contains a contemporaneous B-mode image and a short-ensemble color-flow (CFM) slow-time block with $N \in \{5, 9, 17\}$ pulses. Leonov *et al.* explicitly report PRF for some acquisition families, including the transverse linear-probe Gammex acquisition, but also note that PRF information is not consistently preserved across the database. We therefore fix PRF once per sequence family, using the source-paper value where it is explicitly documented and otherwise recording a sequence-level override in bundle metadata. Band-discretization sanity checks flag invalid Pf/Pa discretizations explicitly. Decoding/QC details are provided in Supplementary Material (PRF provenance and decoder sanity); reproduction commands and manifests are provided in Supplementary Material. Additional artifact-heavy phantom sanity checks are kept in the methods companion.

B-mode-only structural tube masks (independent H_1/H_0). For the Gammex flow phantom sequences we derive evaluation masks from B-mode structure only (no Doppler or matched-subspace outputs). Using acquisition geometry metadata that specifies the CFM region-of-interest (ROI) within the B-mode grid (beam and sample offsets and any beam subsampling), we map the B-mode

ROI onto the CFM grid and segment the tube lumen on a median log-envelope reference image. This yields (i) a conservative lumen (flow) mask, eroded to avoid wall pixels and (ii) a background mask that excludes a guard band around the lumen and bright reflectors. Mask provenance and QC metrics are recorded with each experiment; Figure 9 shows representative overlays. These structural labels support a defensible phantom ROC.

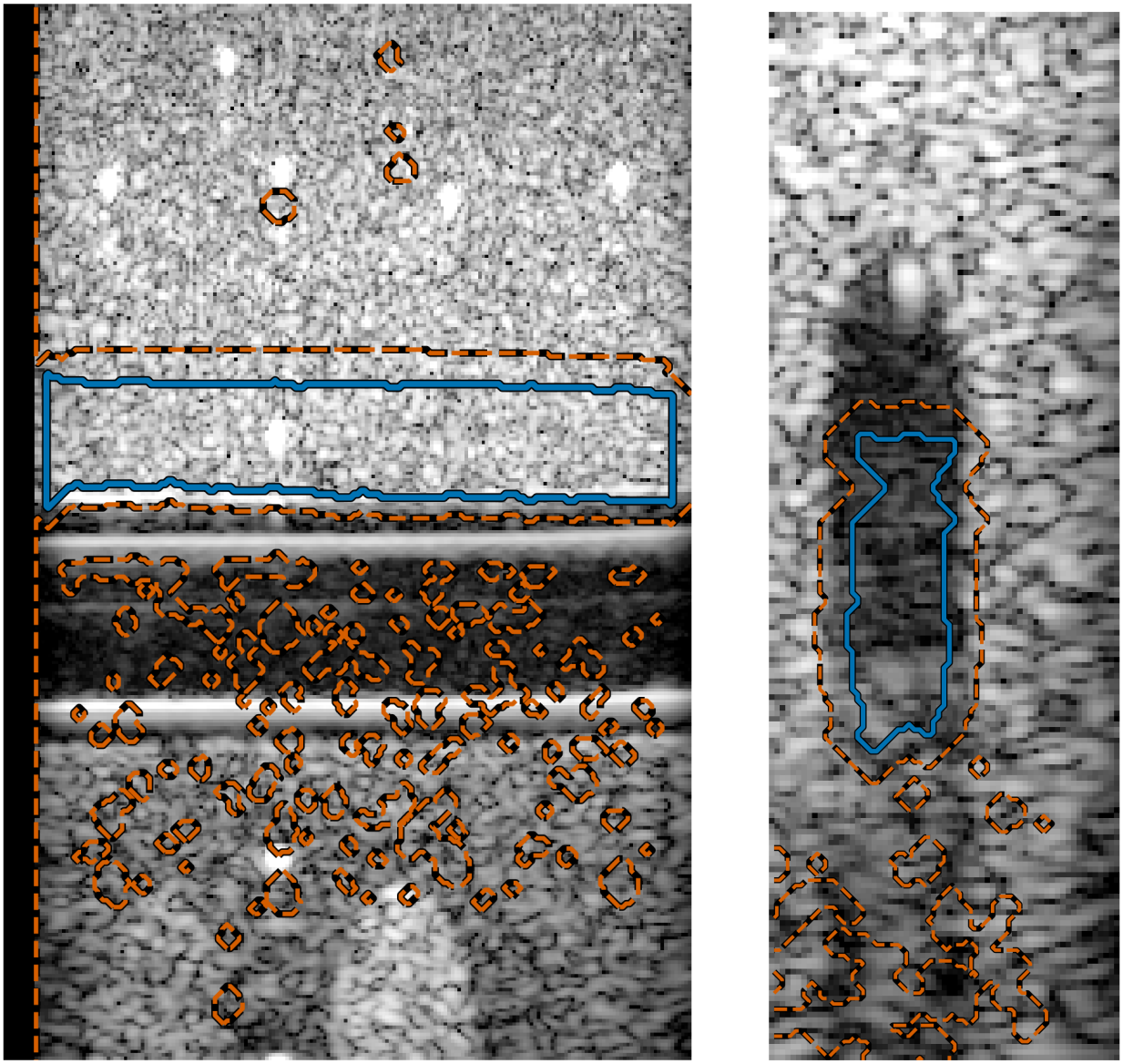

Figure 9: Gammex flow phantom structural evaluation masks derived from B-mode only. Left: along-linear view; right: across-linear view. The lumen mask is intentionally eroded to avoid wall/partial-volume pixels, and the background mask excludes a guard band around the lumen and bright reflectors. Images show the CFM-aligned B-mode ROI with contour overlays of the tube lumen (flow) mask (blue) and background mask (orange, dashed) used for structural-label ROC reporting. Reproducibility details are provided in Supplementary Material.

Frozen short-ensemble Doppler profile and baseline fairness scores. For the linear-probe Gammex sequences used here ($N = 17$ shots; PRF values listed in Supplementary Material), we use a prespecified short-ensemble operating profile: baseline SVD band-pass keep range $[2, 17]$, detector aperture $L_t = 16$, tiles 8×8, a Tyler-type robust covariance estimator, and diagonal

loading $\lambda = 0.07$. To keep the profile auditable while avoiding sample-support limitations on small ROIs and excessive overlap/mixing on large ROIs, we use a deterministic geometry-only stride policy: choose the largest tile stride ≤ 6 such that the implied tile count is ≥ 500. On the two Gammex views this yields stride 6 (along; large ROI) and stride 4 (across; small ROI). Band definitions are set to $P_f = [150, 450]$ Hz and $P_a = [700, 1200]$ Hz. For baseline context on short ensembles, we also report conventional comparators derived from the baseline-filtered cube (power Doppler, log power Doppler, and Kasai lag-1 power). Reproducibility details are provided in Supplementary Material.

Structural-label operating points (Gammex). The same strict-tail geometry of Figure 4 appears in the short-ensemble Gammex phantom, where residual twinkling structure dominates the upper background tail. We therefore pool pixel scores across cine frames and report low-FPR operating points using right-tail thresholding on pooled background pixels, then evaluate TPR on pooled lumen pixels. In these short-ensemble Doppler acquisitions, the background set includes strong artifact pixels whose right-tail scores dominate conventional baselines; consequently, low-FPR thresholds at $\alpha \leq 10^{-3}$ can lie above essentially all lumen pixels, yielding numerically zero baseline TPR at the reported operating points. Figure 10 shows one representative pooled baseline comparator. To avoid treating pooled pixels as independent samples, we report 95% confidence intervals via a frame-level bootstrap: we resample cine frames as the unit of variation while holding the pooled threshold fixed. Table 12 summarizes a broader same-residual score comparison for the two Gammex views under those same settings: power Doppler and Kasai on the baseline residual, plus three detector-family score variants evaluated on that identical residual cube (whitened power, the fixed matched-subspace detector, and the fully whitened matched-subspace detector). For compactness we omit log-PD from the main table because it is monotone-equivalent to PD here and numerically identical at the reported operating points; the full pooled summaries remain available in Supplementary Material. The practical relevance is already visible at a conventional operating point. On the representative Figure 2 frame at matched background FPR 10^{-2}, the baseline thresholded map contains 884 suprathreshold pixels and none lie inside the structural lumen mask, whereas the matched-subspace score recovers 586 lumen pixels while holding the same background allowance. Table 12 now reports that relaxed $\alpha = 10^{-2}$ point alongside the stricter tail operating points, so the phantom result can be read both as a practitioner-facing map-quality summary and as a stress test. The same-residual phantom comparison shows that the baseline residual still contains recoverable lumen signal, because changing only the downstream score restores nonzero sensitivity. In this short-ensemble regime, local whitening is the dominant ingredient, and total whitened slow-time power is stronger than the matched-subspace ratio on both views, especially across-view. Gammex therefore supports localized covariance-aware downstream scoring, but not the claim that the matched-subspace ratio is the preferred score in every phantom regime. The view asymmetry still matters: the across view provides the clearest structural-label phantom evidence, whereas the along view remains a harsher artifact-tail stress test under this short-ensemble profile. A small untuned sanity audit in Supplementary Material gives similar values: on the first 20 cine frames, applying the matched-subspace detector directly to raw registered IQ changes TPR@10^{-3} only from 3.48×10^{-4} to 2.61×10^{-4} on the along view and from 0.202 to 0.205 on the across view; perturbing the frozen along-view PRF from 2000 to 3000 Hz leaves TPR@10^{-3} unchanged at 3.48×10^{-4} in that limited audit. The phantom section is therefore an artifact-tail stress test under a prespecified short-ensemble profile, not proof that every conventional head is universally

zero under any plausible PRF or residualizer.

	Along (linear17 @ PRF 2500)				Across (linear17 @ PRF 2500)			
Method / score	TPR@$\alpha = 10^{-2}$	TPR@$\alpha = 10^{-4}$	TPR@$\alpha = 3\times10^{-4}$	TPR@$\alpha = 10^{-3}$	TPR@$\alpha = 10^{-2}$	TPR@$\alpha = 10^{-4}$	TPR@$\alpha = 3\times10^{-4}$	TPR@$\alpha = 10^{-3}$
Baseline (power Doppler)	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]
Baseline (Kasai lag-1 power)	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]
Detector ablation (whitened power)	0.4116 [0.4078,0.4155]	0.0012 [0.0011,0.0014]	0.0119 [0.0114,0.0124]	0.0532 [0.0521,0.0543]	0.7234 [0.7206,0.7262]	0.4820 [0.4780,0.4863]	0.5155 [0.5115,0.5191]	0.5813 [0.5783,0.5843]
Detector ablation (unwhitened ratio)	0.00031 [0.00022,0.00041]	0.00000 [0.00000,0.00001]	0.00000 [0.00000,0.00001]	0.00001 [0.00000,0.00002]	0.1095 [0.1040,0.1148]	0.0574 [0.0533,0.0611]	0.0602 [0.0562,0.0643]	0.0653 [0.0610,0.0695]
Fully whitened matched-subspace detector	0.0942 [0.0926,0.0958]	0.00000 [0.00000,0.00000]	0.00003 [0.00000,0.00007]	0.00031 [0.00020,0.00043]	0.6891 [0.6846,0.6939]	0.0491 [0.0469,0.0513]	0.0998 [0.0970,0.1027]	0.2237 [0.2197,0.2276]

Table 12: Gammex flow phantom structural-label operating points using B-mode-derived structural masks. All scores are right-tail (higher indicates more flow evidence). This is a same-residual detector-head comparison: every row is evaluated on the identical baseline SVD band-pass residual cube, changing only the detector head. The baseline rows are power Doppler and Kasai lag-1 power on that residual; the detector ablations replace the fully whitened matched-subspace detector with either total whitened slow-time power (no Doppler band partition) or the same flow-band matched-subspace ratio without covariance whitening ($R = I$). The table includes $\alpha = 10^{-2}$ as a practitioner-facing relaxed operating point together with the stricter $\alpha \leq 10^{-3}$ tail regime used for stress testing. Thresholds are chosen per method on pooled background pixels to match each target FPR and evaluated on pooled lumen pixels. Brackets show 95% frame-bootstrap CIs ($n = 2000$ resamples over cine frames) at the fixed pooled threshold. Pooled background counts are $n_{\mathrm{bg}} = 3.85 \times 10^6$ for the along view ($fpr_{\min} \approx 2.6 \times 10^{-7}$) and $n_{\mathrm{bg}} = 1.97 \times 10^6$ for the across view ($fpr_{\min} \approx 5.1 \times 10^{-7}$). Reproducibility details are provided in Supplementary Material.

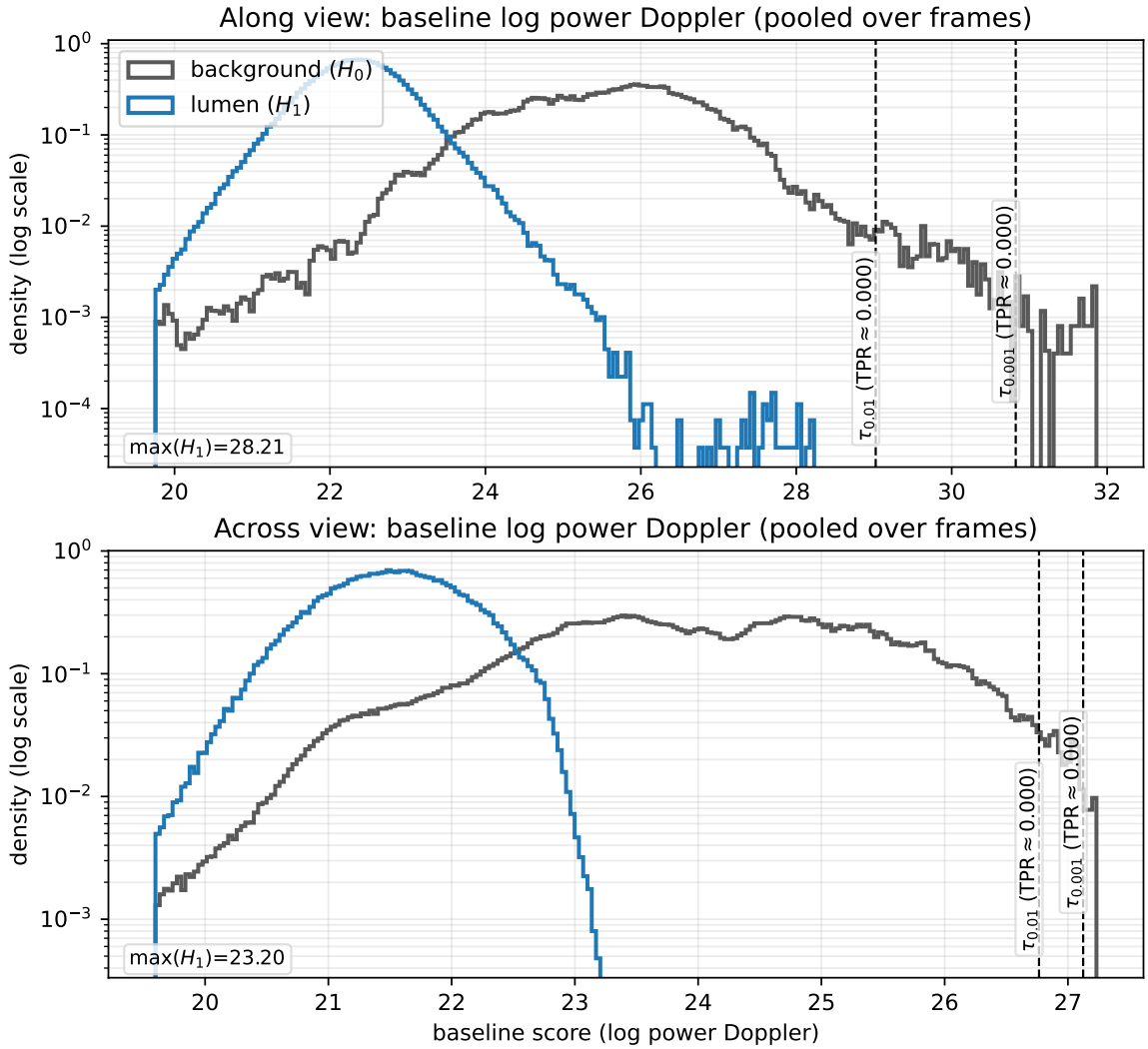

Figure 10: Why conventional baselines can have zero TPR at ultra-low FPR on Gammex short-ensemble data. The plot shows pooled score distributions for the baseline log power Doppler comparator on lumen (H_1; structural tube mask) versus background (H_0; structural background mask), for both along and across views. Vertical lines indicate background-calibrated right-tail thresholds τ_α at $\alpha \in \{10^{-2}, 10^{-3}\}$; $\alpha = 10^{-2}$ is shown here for context because the reported operating points are $\alpha \leq 10^{-3}$. On both views the background tail threshold lies above $\max(H_1)$, so $\mathrm{TPR}(\alpha) = \mathbb{P}\{S \geq \tau_\alpha \mid H_1\}$ is numerically zero at these tail operating points.

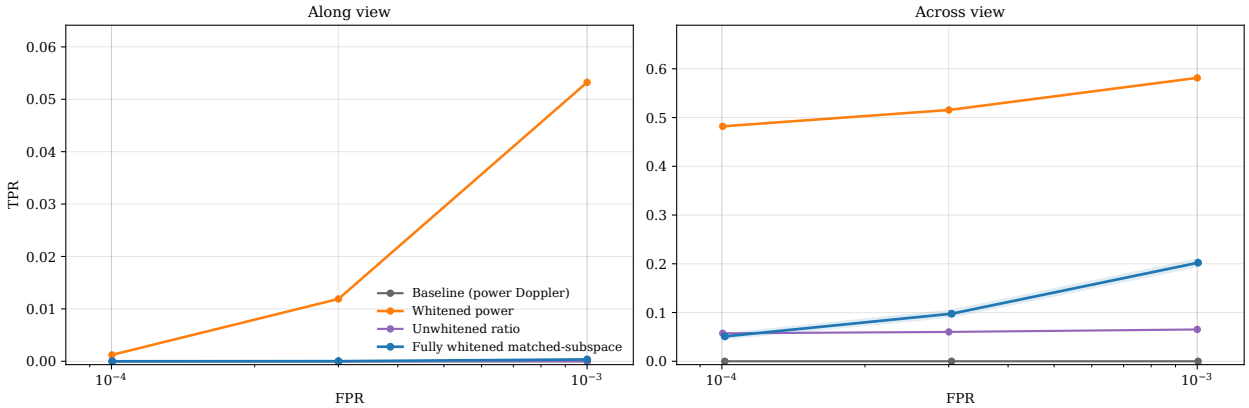

Figure 11: Gammex flow phantom same-residual score comparison in the strict low-FPR region used for the main phantom table. Curves connect the three reported operating points for baseline power Doppler, whitened power, the fixed matched-subspace detector, and the fully whitened matched-subspace detector. The two panels use panel-specific TPR ranges so that the operating-point differences are readable.

Additional Gammex sanity checks are provided in the Supplementary Material.

Within-ensemble motion ladders. A labeled within-ensemble motion ladder on the along-view Gammex cine is included in the methods companion. It is directionally consistent with the main phantom result and is useful as a robustness check, but the central phantom claim in the main preprint is the same-residual structural ROC comparison above.

5.4 ULM real-IQ audits

Dataset and loader. We use a real kHz IQ dataset released for ultrasound localization microscopy benchmarking [37, 38] and distributed as HDF5 blocks with accompanying acquisition metadata. Each block contains a dataset `iq` of shape $(2, H, W, T) = (2, 256, 180, 800)$ float32 representing I/Q over $T = 800$ slow-time frames. We load blocks on demand and return complex IQ as a cube of shape (T, H, W) (reproduction details in Supplementary Material).

One-time label-free baseline selection. For ULM we select a single MC–SVD energy-fraction cutoff by a one-time label-free sweep on blocks 1–3 over frames 0:128, and then freeze it for all subsequent ULM motion experiments. The sweep uses baseline-only runs and relies on label-free integrity and stability criteria (no task labels); it yields a frozen energy fraction $e = 0.975$ (full sweep details in Supplementary Material). The later PALA-backed structural audit and block-split threshold-transfer analysis use the same open dataset and therefore overlap in source blocks, but this one-time sweep used no localization-derived structural labels and fixed only the baseline MC–SVD energy fraction before those analyses were run.

5.4.1 Localization-derived structural audit

ULM-CoreShell. To obtain a more anatomical structural readout on the same open real-IQ data, we use the published PALA rat-brain ULM example as an external vascular reference derived from

microbubble localization microscopy rather than from the Doppler scores being evaluated [37, 38]. We first reproduce the corresponding short-horizon PALA PowDop rendering from the Zenodo IQ blocks using the published odd-block selection and SVD cutoff, register that published PowDop crop back to the Zenodo IQ field of view, and then downsample the published super-resolution MatOut vascular rendering onto the short-window evaluation lattice. From that registered localization map we derive a conservative vessel-core versus perivascular-shell label pair: the vessel mask is a peak-based core mask, and the background mask is the surrounding shell, so the endpoint asks whether a short-window detector concentrates score on the vascular core rather than the immediately adjacent parenchyma. That one-time PALA-to-IQ registration is part of the label construction only. For the short-window detector itself we keep frame-to-frame registration disabled in the frozen audit, because re-enabling it changes the structural ROC only below the fourth decimal place while adding substantial latency.

On this endpoint, Table 2 reports PD, Kasai, the fixed head, the adaptive head, and the fully whitened variant on the same residual windows and the same localization-derived vessel-core versus shell masks. The fixed and adaptive heads yield only modest pooled-ROC gains over PD and Kasai, and adaptive is nearly indistinguishable from fixed under this profile. The fully whitened variant is the only head that improves pooled AUC, strict-tail sensitivity, and shell FPR at matched 70% vessel-core recall at once: AUC 0.530 [0.524, 0.537], TPR@10^{-3} 0.004 [0.003, 0.006], and shell FPR 0.676 [0.665, 0.687], compared with AUC 0.514/0.514, TPR@10^{-3} 0.000/0.000, and shell FPR 0.703/0.702 for the fixed/adaptive heads and AUC 0.504/0.503, TPR@10^{-3} 0.000/0.000, and shell FPR 0.698/0.701 for PD/Kasai. ULM-CoreShell is therefore a real-IQ regime in which the fully whitened variant is preferred rather than one in which the default fixed head carries over unchanged. In practical terms, the paper supports a conservative rule: use the fixed variant as the default when no regime-specific validation is available, and treat the fully whitened variant as a narrower option for settings that show persistent structured residual clutter or reward vascular-core concentration. The current adaptive switch is a plausible mechanism for that transition, but in ULM-CoreShell it tracks the fixed head rather than the fully whitened variant, so it should not yet be read as a validated prospective selector. That endpoint-specific behavior is plausible: the ULM vessel-core endpoint rewards concentration on the localization-derived vascular core relative to the adjacent shell, whereas SIMUS-Struct rewards strong fixed downstream robustness against broader structured nuisance.

The pooled advantage is not driven by one favorable block. Relative to the fixed head, the fully whitened variant exceeds AUC on 47/70 audited windows and all 10/10 blocks, lowers shell FPR at matched 70% core recall on 47/70 windows and 9/10 blocks, and yields nonzero TPR@10^{-3} in 45/70 windows versus 10/70 for the fixed and adaptive heads (Table 13). The same pattern holds against the adaptive head: 46/70 AUC windows, 47/70 shell-FPR windows, and 10/10 blocks. Against baseline power Doppler, the fully whitened variant also wins AUC on all 10 audited blocks, and an exact paired sign-flip test on the block-mean AUC difference gives $p = 0.00195$. Figure 1 gives the visual counterpart to this result by showing a representative same-window comparison between baseline power Doppler and the fully whitened variant. On an RTX 4080 SUPER, the same 64-frame no-registration configuration runs at 41.3 ms steady-state total for a 64 ms acquisition budget under the reported replay conditions. That is replay-feasibility evidence only, not scanner integration or OR-ready deployment. The labels come from a published localization-derived vascular rendering on the same acquisition rather than from an independent modality or task-based ground truth, so this remains a structural rather than functional result.

To reduce overlap between localization-derived label construction and scored windows, we also ran a stricter odd-block-reference / even-block-evaluation variant under the same frozen 64-frame profile. The family ordering remained directionally consistent: the fully whitened variant still led on pooled AUC and strict-tail sensitivity, though with smaller margins than in the main same-acquisition audit. This stricter split reduces, but does not eliminate, the same-acquisition overlap limitation, and the summary is provided in the Supplementary Material. A supplementary falsification control points in the same direction: support-matched randomization of the vessel-core and shell masks collapses the structural audit toward chance, while a large cyclic translation destroys the fully whitened ordering rather than reproducing it.

Comparison	AUC wins (windows / blocks)	shell-FPR wins @70% core recall (windows / blocks)	nonzero TPR@10^{-3} windows
Fully whitened variant vs fixed head	47/70 / 10/10	47/70 / 9/10	45/70 vs 10/70
Fully whitened variant vs adaptive head	46/70 / 10/10	47/70 / 10/10	45/70 vs 10/70

Table 13: Window-level consistency summary for the PALA-backed ULM structural audit. Each row compares the prespecified fully whitened variant against another detector head on the same frozen 64-frame MC–SVD residual cube and the same localization-derived vessel-core versus shell labels. The window/block entries report how often the fully whitened variant has higher vessel-core AUC or lower shell false-positive rate at matched 70% vessel-core recall.

A second transfer analysis uses the same structural endpoint but removes per-window retuning. We split the 10 audited blocks into two complementary 5-block calibration/evaluation halves, learn a single shell-calibrated threshold on one half, and apply it unchanged to the other half at nominal $\alpha = 10^{-3}$. Table 14 shows the result. Under that fixed threshold transfer, PD and Kasai collapse to zero held-out vessel-core recall, whereas the fully whitened matched-subspace variant retains a small but nonzero held-out core TPR of 0.002 [0.000, 0.004] at a realized shell FPR of 0.001 [0.000, 0.002]. More concretely, it is the only score with any positive held-out core detections at all, firing on 6 of the 20 held-out windows versus 0 of 20 for PD and Kasai. The effect is still modest, but it shows that the structural claim does not depend entirely on per-window threshold readjustment.

Score	core TPR	realized shell FPR
Baseline (power Doppler)	0.000 [0.000,0.000]	0.001 [0.000,0.002]
Baseline (Kasai lag-1 magnitude)	0.000 [0.000,0.000]	0.001 [0.000,0.002]
Whitened matched-subspace variant	0.002 [0.000,0.004]	0.001 [0.000,0.002]

Table 14: Fixed-calibration transfer on the PALA-backed ULM structural audit at nominal $\alpha = 10^{-3}$. A threshold is learned once from pooled perivascular-shell negatives on one 5-block calibration bank and then applied unchanged to the complementary 5 held-out blocks; the reported values pool the two complementary block splits and show window-level means with 95% bootstrap CIs over held-out windows. This is therefore a genuine non-retrospective threshold-transfer check on real in-vivo IQ rather than per-block retuning.

5.4.2 Motion robustness audit

Companion motion audit. A companion-only motion-injection audit on ULM-CoreShell and Shin-ProxyMotion provides secondary robustness evidence under controlled residual-motion stress. In those audits, matched-subspace score maps are more stable than the baseline maps, but these

results are robustness corroboration rather than the structural efficacy claim that anchors the main real-IQ section.

Shin-ProxyMotion. To avoid making the motion-robustness story depend on a single open dataset, we also summarize the strongest existing Shin rat-brain motion-stress result from the supplement [35,36]. Under a prespecified low-frequency operating setting and a controlled residual-motion stress test, the matched-subspace score map is more stable than the MC–SVD baseline map across all three audited windows, and the matched background-tail operating point yields a nonzero flow-proxy hit rate only for the matched-subspace detector.

Window	self-correlation base→matched-subspace	flow hit at bg tail 10^{-3} base→matched-subspace	clip fraction matched-subspace>base (correlation)	clip fraction matched-subspace>base (hit rate)
0:128	0.188 → 0.454	0.000 → 0.041	1.0	1.0
64:192	0.186 → 0.399	0.000 → 0.040	0.9	1.0
122:250	0.182 → 0.364	0.000 → 0.041	0.8	1.0

Table 15: Compact Shin rat-brain open real-IQ corroboration under controlled residual-motion stress (10 clips; 2 px amplitude; rigid registration enabled). "Self-correlation" is the Pearson correlation between each method's score map and its own no-motion reference after alignment and a boundary crop. "Flow-proxy hit rate" is evaluated at a matched background-proxy tail rate of 10^{-3} under a fixed, method-independent flow-proxy mask.

A separate fixed-threshold check on Shin asks a different question. Using the same proxy masks, we learn one background-proxy threshold from five clips and apply it unchanged to the complementary five clips across all three audited windows at nominal $\alpha = 10^{-3}$. Table 16 shows that this cross-clip proxy transfer collapses for all three scores: PD, Kasai, and the fixed matched-subspace detector each fall to zero held-out flow-proxy hit rate at realized background-proxy FPR near target.

Score	flow-proxy hit rate	realized bg-proxy FPR
Baseline (power Doppler)	0.000 [0.000,0.000]	0.001 [0.001,0.001]
Baseline (Kasai lag-1 magnitude)	0.000 [0.000,0.000]	0.001 [0.001,0.001]
Fixed matched-subspace detector	0.000 [0.000,0.000]	0.001 [0.001,0.002]

Table 16: Fixed-calibration transfer on the Shin rat-brain real-IQ proxy audit at nominal $\alpha = 10^{-3}$. A threshold is learned once from pooled background-proxy negatives on one 5-clip calibration bank and then applied unchanged to the complementary 5 held-out clips across all three audited windows per clip. Reported values pool the two complementary clip splits and show scenario-level means with 95% bootstrap CIs. Because Shin does not provide independent task labels, this is a secondary proxy-based deployment check rather than a task ROC result.

The real-data results span four roles: Gammex-Struct provides structurally labeled phantom evidence, ULM-CoreShell adds a structural check using a localization-derived vascular reference plus block-split threshold transfer, ULM-Motion provides open real-IQ robustness evidence, and Shin-ProxyMotion provides a second open real-IQ corroboration under controlled residual-motion stress plus a negative cross-clip proxy-transfer check. A compact same-cube score audit across

those real-IQ bundles is kept in the Supplementary Material, together with additional label-free audits such as whole-brain PD-only retrospectives, Shin contract summaries, and extended motion variants.

5.5 Computational cost and replay feasibility

We also report representative runtimes for the final detector family evaluated in this manuscript rather than for a whitening-heavy research configuration. In this evaluation, the labeled-brain stress tests use the fixed matched-subspace default with conditional execution, the Shin real-IQ evaluation uses the adaptive variant, and Gammex uses the short-ensemble raw-baseline plus fixed matched-subspace head. Conditional execution changes the decision rule by deciding which tiles receive the adaptive detector, so it is reported as part of the detector configuration rather than as a free acceleration of the same method. Unless otherwise stated, conditional-execution reporting uses the same leakage-controlled, disjoint-window proxy discipline as the detection results.

Latency is measured by replaying multiple windows or frames in a single GPU process and reporting *steady-state* means over windows/frames 2..N. The first replay window/frame is excluded because it includes one-time startup overheads. We time the baseline and detector stages separately, excluding disk I/O, and "Total" reports baseline plus detector wall time. "Budget" denotes the acquired slow-time duration of the corresponding ensemble: $64/1500 \approx 42.7$ ms for the brain windows, $128/1000 = 128$ ms for Shin, and $17/2500 = 6.8$ ms for the Gammex cines. The reported real-time factor (RTF) is therefore Total/Budget.

Scenario	Baseline (s)	Detector (s)	Total (s)	Budget (s)	RTF
Open-skull surface-artifact stress test (64-frame window)	0.024	0.012	0.037	0.043	0.86
Alias-dominant positive control (64-frame window)	0.025	0.012	0.038	0.043	0.88
Structured-clutter leakage stress test (64-frame window)	0.026	0.012	0.037	0.043	0.87
Shin rat-brain real-IQ evaluation (128-frame window)	0.006	0.102	0.108	0.128	0.84
Gammex flow phantom, along view (17-shot cine frame)	0.000	0.006	0.006	0.007	0.88
Gammex flow phantom, across view (17-shot cine frame)	0.000	0.004	0.004	0.007	0.64

Table 17: Steady-state latency summary on an NVIDIA GeForce RTX 4080 SUPER 16GB for the final detector family evaluated in this manuscript. Times are means over windows/frames 2..N, exclude disk I/O, and compare baseline and detector stages under the reported detector configuration rather than under an earlier whitening-heavy timing setup.

Table 17 shows that the replay picture is substantially better in the final reported configurations than in a dense fully whitened setup. All three labeled-brain stress tests run below the full acquisition-PRF budget (RTF $\approx$ 0.86–0.88), the Shin rat-brain real-IQ evaluation also remains below budget (RTF $\approx$ 0.84), and both Gammex views run below the 17-shot cine-frame budget (RTF $\approx$ 0.88 along-view and $\approx$ 0.64 across-view).[1] These timings support workstation replay feasibility

[1]Latency is reported from steady-state replay (windows/frames 2..N) and excludes startup overhead and disk

for the reported configurations, but not scanner-integrated or cross-session deployment.

A leakage ablation confirms that conditional execution can inflate apparent performance under same-window proxy gating, while leakage-controlled gating yields a small TPR decrease (order 10^{-3}) at the target FPRs. The full open-skull leakage table with bootstrap intervals is provided in the Supplementary Material.

6 Discussion

What the detector family adds

Post-beamformed fUS benefits from a better detector, not only from better residualization. Ordinary clutter suppression can still leave structured right-tail false detections, and the central result of this paper is that changing only the downstream score head can materially change nuisance control on the same residual cube. The mechanism is regime-dependent: matched-subspace geometry is the main driver on the labeled-brain stress tests and the held-out SIMUS benchmark, whereas local slow-time whitening becomes more important in short-ensemble or structure-concentration regimes. Gammex-Struct is the clearest counterexample to a universal matched-subspace claim: on the same residual phantom data, whitened power is stronger than the matched-subspace ratio on both views. The paper therefore supports a detector family with a clear fixed default and narrower whitened variants rather than one head or one mechanism that wins everywhere.

For a practitioner, the immediate implication is that the same SVD residual already used in existing pipelines can yield a more trustworthy score map by changing only the final scoring step. The fixed head requires no additional training data and no retuning of the upstream clutter filter. On the held-out benchmark, it eliminates all nuisance detections at matched 50% target-flow recall on the same residual where power Doppler passes 97–100% of nuisance. That is not a marginal image-quality change; it is the difference between a map that can be thresholded with confidence and one that still requires visual inspection to separate artifact from signal. Whether that difference transfers fully to real *in vivo* task data remains the central open question.

The comparator set is intentionally residualizer-heavy rather than learning-heavy. Learned clutter filters remain important related work, but they address residual formation rather than the downstream detector-head question isolated here. Extending the same-residual comparisons in this paper to trained detector heads or learned residualizers would require a separate supervised training protocol and is left for future work.

The strict-tail operating points in the labeled experiments are calibrated using known negative sets, but a scanner does not have frame-by-frame labeled negatives at runtime. Here that tail calibration is an *evaluation protocol*, not the intended clinical operating rule. The transfer checks show some non-retrospective signal on SIMUS-Struct and ULM-CoreShell, while the Shin proxy transfer remains negative, so a practical thresholding workflow on raw *in vivo* IQ remains unestablished.

Real-data scope and clinical interpretation

Most public *in vivo* fUS releases do not preserve the raw high-PRF complex IQ and metadata needed for independent detector evaluation after beamforming. For that reason, the strongest efficacy evidence in this paper still comes from labeled simulations and structurally labeled phantom

I/O. Table 17 reports the final detector family used in the manuscript rather than a single worst-case fully whitened configuration applied everywhere.

data, while the open real-IQ sections are narrower robustness and structural audits. The ULM-CoreShell result is directionally positive but modest, and only the fully whitened variant shows a clear advantage there. The real-data contribution is therefore improved vascular and nuisance discrimination at the detector stage, not biological inference or real-data functional task validation.

A stricter disjoint-source odd/even block audit remained directionally consistent but weaker, and a support-matched permutation control collapsed the structural audit toward chance. Together these checks reduce the simplest same-acquisition leakage objections, but they do not turn the result into independent multimodal or cross-session validation.

The simulation-to-reality gap should calibrate expectations: the held-out synthetic benchmark determines the default detector choice much more cleanly than the current real-IQ audits. What is still missing is a standardized open benchmark with repeated sessions and independent anatomical or task anchors for post-beamformed detector evaluation [8, 22–25].

Recommended detector variant by regime

The practical deployment rule is conservative. Start with the fixed head. If runtime telemetry from the adaptive branch shows persistently elevated guard-band clutter, non-negligible promotion fractions, and acceptable latency margin, the fully whitened variant is justified; otherwise keep the fixed default. The current telemetry is useful as label-free clutter monitoring, but it does not yet recover the ULM switch prospectively, so the adaptive branch should not be read as a validated universal selector.

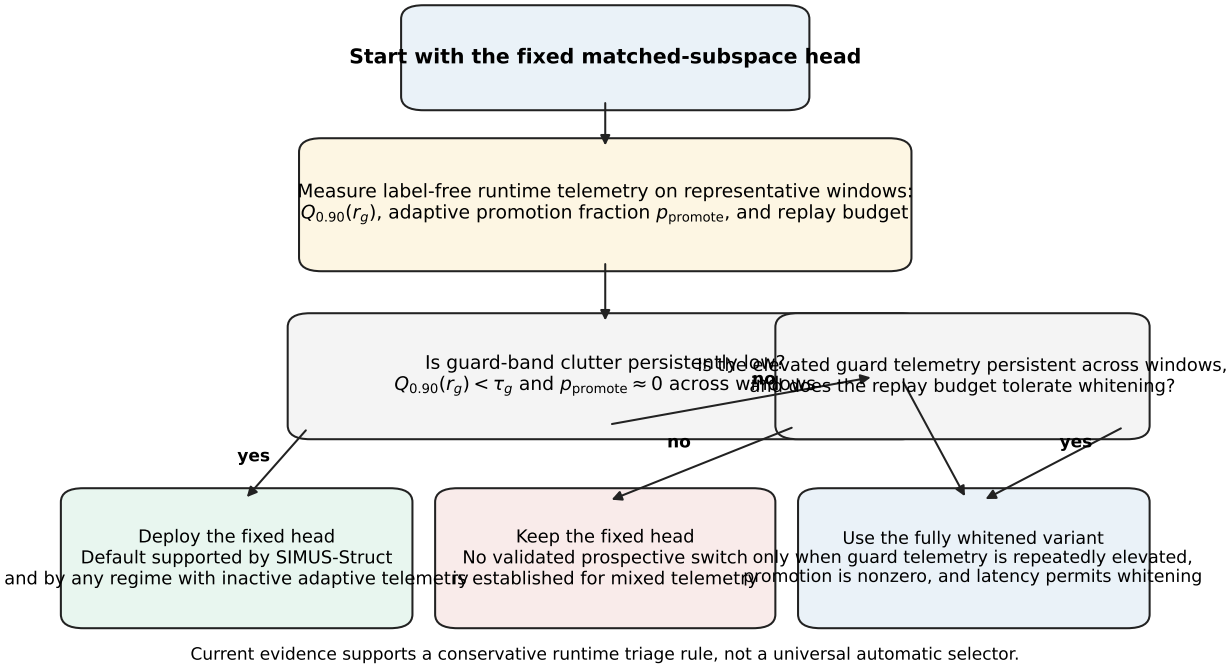

Figure 12: Conservative label-free deployment rule for choosing between the fixed and fully whitened detector variants. The runtime telemetry is already produced by the adaptive branch: r_g is the tile-level guard-band energy fraction, $Q_{0.90}(r_g)$ is its upper-tail summary across a representative window, and p_{promote} is the fraction of tiles that would be promoted onto the whitened branch by the frozen adaptive rule. The flowchart encodes the present practical conclusion: fixed is the default, fully whitened is reserved for repeated guard-dominant windows with nonzero promotion and sufficient latency budget, and mixed telemetry is not enough to justify an automatic switch. This is a conservative deployment heuristic from the current evidence, not a validated universal selector.

7 Conclusion and future work

We presented a localized matched-subspace detector framework for post-beamformed fUS and ultrafast Doppler data, centered on a band-limited fixed head and extended by an adaptive whitening switch plus a fully whitened variant. Across the evaluated labeled simulations, SIMUS-Struct, and one structurally labeled phantom regime, this detector family often improved low-FPR nuisance control relative to PD/Kasai-style downstream scores, but the preferred downstream head was regime-dependent. Across those tiers, the fixed band-limited head remains the structural default, while the adaptive and fully whitened variants help in selected clutter-heavy regimes rather than serving as universal replacements. The phantom result should be read accordingly: it supports localized covariance-aware downstream scoring, but not a universal matched-subspace mechanism.

On one open real-IQ rat-brain dataset, the same-acquisition ULM structural audit shows only modest gains for the fixed and adaptive heads, while the fully whitened variant is the only head that clearly improves pooled AUC, strict-tail sensitivity, and shell suppression on the vessel-core

versus perivascular-shell endpoint. A stricter disjoint-source odd/even block variant preserved that ordering with attenuated margins, so the real-IQ structural signal is not wholly dependent on block overlap inside the same acquisition. Fixed-threshold transfer remains only partly established: SIMUS-Struct and the ULM structural audit retain non-retrospective signal, while the Shin proxy transfer stays negative. The method is therefore positioned as a detector-stage contribution for post-beamformed fUS pipelines, not as completed real-data functional validation. The main next external test is prospective labeled *in vivo* validation on raw high-PRF IQ with independent anatomical or task anchors and a practical non-retrospective thresholding workflow.

More broadly, the same-residual evaluation framework used here, freezing the clutter-suppressed cube and comparing only the downstream score, provides a general protocol for benchmarking post-beamformed fUS detection methods and could serve as a standardized evaluation layer as the field moves toward open reproducible pipelines.

Data and code availability

All datasets used in this work are either publicly available (as cited) or derived deterministically from those sources. All code used to generate the reported tables and figures is versioned; the exact commit hash and release information are recorded in the reproducibility manifest (Supplementary Material). A minimal CUDA-enabled software environment and container recipe are provided in the repository. End-to-end regeneration commands for the reported figures and tables are collected in the repository and summarized in the reproducibility appendix.

Code repository. The reference implementation and reproduction scripts are available at `https://github.com/ArthurShune/fus-detectors`. The canonical commit hash and release information are recorded in the reproducibility manifest (Supplementary Material).

References

[1] Imbault, M., et al. Intraoperative functional ultrasound imaging of human brain activity. *Scientific Reports* 7, 7304 (2017).

[2] Macé, E., Montaldo, G., Pernot, M., Cohen, I., and Tanter, M. Functional ultrasound imaging of the brain. *NeuroImage* 54(4), 2720–2731 (2011).

[3] Urban, A., et al. In vivo whole brain microvascular imaging in mice using ultrafast ultrasound. *EBioMedicine* 76, 103861 (2022).

[4] Alexandrov, A. V., et al. Velocity measurements in the middle cerebral arteries of healthy adults by transcranial Doppler. *Journal of Neuroimaging* 1, 166–172 (1991).

[5] Alexandrov, A. V., et al. Optimal values of flow velocity on transcranial Doppler in patients with intracranial arterial stenosis. *Stroke* 33, 1757–1761 (2002).

[6] Macé, E., et al. Functional ultrasound imaging of the brain: theory and basic principles. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* 60(3), 492–506 (2013).

[7] Rabut, C., et al. Functional ultrasound imaging of human brain activity through the skull. *Science Translational Medicine* 12(572), eaav4392 (2020).

[8] Rabut, C., Norman, S. L., Griggs, W. S., Russin, J. J., Jann, K., et al. Functional ultrasound imaging of human brain activity through an acoustically transparent cranial window. *Science Translational Medicine* 16, eadj3143 (2024).

[9] Deffieux, T., Demené, C., and Tanter, M. Functional ultrasound imaging: A new imaging modality for neuroscience. *Neuroscience* 474, 110–121 (2021).

[10] Demeulenaere, O., Bertolo, A., Pezet, S., Osmanski, B.-F., Tanter, M., and Deffieux, T. In vivo whole brain microvascular imaging in mice using transcranial 3D ultrasound localization microscopy. *EBioMedicine* 79, 103995 (2022).

[11] Demené, C., et al. Spatiotemporal clutter filtering of ultrafast ultrasound data highly increases Doppler and fUltrasound sensitivity. *IEEE Transactions on Medical Imaging* (2015). DOI: `https://doi.org/10.1109/TMI.2015.2428634`.

[12] Baranger, J., et al. Adaptive spatiotemporal SVD clutter filtering for ultrafast Doppler imaging using similarity of spatial singular vectors. *IEEE Transactions on Medical Imaging* (2018). DOI: `https://doi.org/10.1109/TMI.2018.2789499`.

[13] Yu, A. C. H., and Løvstakken, L. Eigen-based clutter filter design for ultrasound color flow imaging: a review. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* (2010). DOI: `https://doi.org/10.1109/TUFFC.2010.1521`.

[14] Song, P., et al. Ultrasound small vessel imaging with block-wise adaptive local clutter filtering. *IEEE Transactions on Medical Imaging* (2017). DOI: `https://doi.org/10.1109/TMI.2016.2605819`.

[15] Kasai, C., Namekawa, K., Koyano, A., and Omoto, R. Real-time two-dimensional blood flow imaging using an autocorrelation technique. *IEEE Transactions on Sonics and Ultrasonics* 32(3), 458–464 (1985). DOI: `https://doi.org/10.1109/T-SU.1985.31615`.

[16] Chee, A. J. Y., Yiu, B. Y. S., and Yu, A. C. H. A GPU-parallelized eigen-based clutter filter framework for ultrasound color flow imaging. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* (2017). DOI: `https://doi.org/10.1109/TUFFC.2016.2606598`.

[17] Lok, U.-W., Tang, J., Kilic, K., Szabo, T. L., and Boas, D. A. Real time SVD-based clutter filtering using randomized SVD and spatial downsampling for microvessel imaging. *Ultrasonics* 107, 106163 (2020). DOI: `https://doi.org/10.1016/j.ultras.2020.106163`.

[18] Demene, C., et al. Improved color Doppler for cerebral blood flow axial velocity estimation (iCD-fUS). *IEEE Transactions on Medical Imaging* 40(5), 1498–1510 (2021).

[19] Tang, J., Kilic, K., Szabo, T. L., and Boas, D. A. Improved color Doppler for cerebral blood flow axial velocity imaging. *IEEE Transactions on Medical Imaging* 40(2), 758–764 (2021).

[20] Demené, C., et al. Functional ultrasound imaging of brain activity in human newborns. *Science Translational Medicine* 9(411), eaah6756 (2017). DOI: https://doi.org/10.1126/scitranslmed.aah6756.

[21] Baranger, J., et al. Bedside functional monitoring of the dynamic brain connectivity in human neonates. *Nature Communications* 12, 1080 (2021). DOI: https://doi.org/10.1038/s41467-021-21387-x.

[22] Soloukey, S., et al. Mobile human brain imaging using functional ultrasound. *Science Advances* 11(25), eadu9133 (2025). DOI: https://doi.org/10.1126/sciadv.adu9133.

[23] Lambert, T., Brunner, C., Urban, A., et al. Functional ultrasound imaging and neuronal activity: How accurate is the spatiotemporal match? *Imaging Neuroscience* 3, IMAG.a.34 (2025). DOI: https://doi.org/10.1162/IMAG.a.34.

[24] Edelman, B. J., et al. The COMBO window: A chronic cranial implant for multiscale circuit interrogation in mice. *PLOS Biology* 22(6), e3002664 (2024). DOI: https://doi.org/10.1371/journal.pbio.3002664.

[25] Chen, Z., et al. Functional ultrasound imaging of auditory responses in comatose patients. *Research* 8, 0709 (2025). DOI: https://doi.org/10.34133/research.0709.

[26] Nunez-Elizalde, A. O., et al. Neural correlates of blood flow measured by ultrasound. *Neuron* 110(10), 1631–1640.e4 (2022). DOI: https://doi.org/10.1016/j.neuron.2022.02.012.

[27] Aaslid, R., Markwalder, T.-M., and Nornes, H. Noninvasive transcranial Doppler ultrasound recording of flow velocity in basal cerebral arteries. *Journal of Neurosurgery* 57(6), 769–774 (1982).

[28] Gao, S., Lam, W. W.-K., Chan, Y.-H., Liu, J., Wong, K.-P., Leung, T. W.-S., et al. Optimal values of flow velocity on transcranial Doppler in grading middle cerebral artery stenosis in comparison with magnetic resonance angiography. *Journal of Neuroimaging* 12(3), 213–218 (2002).

[29] Estrada, H., Gottschalk, S., Reiss, M., Neuschmelting, V., Razansky, D., et al. Observation of guided acoustic waves in a human skull. *Ultrasound in Medicine & Biology* 44(7), 1438–1447 (2018).

[30] Candès, E. J., Li, X., Ma, Y., and Wright, J. Robust principal component analysis? *Journal of the ACM* 58(3), 11 (2011).

[31] Solomon, O., et al. Deep unfolded robust PCA with application to clutter suppression in ultrasound. *IEEE Transactions on Medical Imaging* 39(4), 1051–1063 (2020). DOI: https://doi.org/10.1109/TMI.2019.2941271.

[32] Lin, Z., Chen, M., and Ma, Y. The augmented Lagrange multiplier method for exact recovery of corrupted low-rank matrices. *arXiv preprint* arXiv:1009.5055 (2010).

[33] Halko, N., Martinsson, P. G., and Tropp, J. A. Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions. *SIAM Review* 53(2), 217–288 (2011).

[34] Soloukey, S., Vincent, A. J. P. E., Smits, M., Slump, C. H., Dirven, C. M. F., et al. Functional ultrasound (fUS) during awake brain surgery: the clinical potential for intra-operative functional and vascular brain mapping. *Frontiers in Neuroscience* 13, 1384 (2019). DOI: https://doi.org/10.3389/fnins.2019.01384.

[35] Shin, Y., et al. Context-Aware Deep Learning Enables High-Efficacy Localization of High Concentration Microbubbles for Super-Resolution Ultrasound Localization Microscopy. Zenodo dataset, DOI 10.5281/zenodo.10711806 (2024).

[36] Shin, Y., et al. Context-Aware Deep Learning Enables High-Efficacy Localization of High Concentration Microbubbles for Super-Resolution Ultrasound Localization Microscopy. *Nature Communications* 15, 3299 (2024). DOI: https://doi.org/10.1038/s41467-024-47154-2.

[37] Chavignon, A., Heiles, B., Hingot, V., Lopez, P., Teston, E., and Couture, O. In vivo rat brain for Ultrasound Localization Microscopy: raw and beamformed data. Zenodo dataset, DOI 10.5281/zenodo.7883227 (2023).

[38] Heiles, B., Chavignon, A., Hingot, V., Lopez, P., Teston, E., and Couture, O. Performance benchmarking of microbubble-localization algorithms for ultrasound localization microscopy. *Nature Biomedical Engineering* (2021). DOI: https://doi.org/10.1038/s41551-021-00824-8.

[39] Errico, C., et al. Ultrafast ultrasound localization microscopy for deep super-resolution vascular imaging. *Nature* 527(7579), 499–502 (2015). DOI: https://doi.org/10.1038/nature16066.

[40] Hingot, V., et al. Subwavelength motion-correction for ultrafast ultrasound localization microscopy. *Ultrasonics* 77, 17–21 (2017). DOI: https://doi.org/10.1016/j.ultras.2017.01.008.

[41] Insana, M. F., Dai, B., Babaei, S., and Abbey, C. K. Combining spatial registration with clutter filtering for power-Doppler imaging in peripheral perfusion applications. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* 69(12), 3243–3254 (2022). DOI: https://doi.org/10.1109/TUFFC.2022.3211469.

[42] Leonov, D. V., Reshetnikov, R. V., Kulberg, N. S., Nasibullina, A. A., and Gromov, A. I. Doppler Twinkling Artifact Observations: an Open-Access Database of Raw Ultrasonic Signals. *Digital Diagnostics* (2021). DOI: https://doi.org/10.17816/DD76511.

[43] Ozgun, K., and Byram, B. C. Multidimensional clutter filtering of aperture domain data for improved blood flow sensitivity. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* 68(8), 2645–2656 (2021). DOI: https://doi.org/10.1109/TUFFC.2021.3073292.

[44] Huang, Z., et al. Adaptive higher-order singular value decomposition clutter filter for ultrafast Doppler imaging of coronary flow under non-negligible tissue motion. *Ultrasonics* (2024). DOI: https://doi.org/10.1016/j.ultras.2024.107307.

[45] Garcia, D. SIMUS: an open-source simulator for medical ultrasound imaging. Part I: theory and examples. *Computer Methods and Programs in Biomedicine* 218, 106726 (2022). DOI: https://doi.org/10.1016/j.cmpb.2022.106726.

[46] Cigier, A., Varray, F., and Garcia, D. SIMUS: an open-source simulator for medical ultrasound imaging. Part II: comparison with four simulators. *Computer Methods and Programs in Biomedicine* 220, 106774 (2022). DOI: https://doi.org/10.1016/j.cmpb.2022.106774.

[47] Urban, A., et al. Real-time imaging of brain activity in freely moving rodents using functional ultrasound. *Nature Methods* 12(9), 873–878 (2015). DOI: https://doi.org/10.1038/nmeth.3482.

[48] Demené, C., et al. 4D microvascular imaging based on ultrafast Doppler tomography. *NeuroImage* 127, 472–483 (2016). DOI: https://doi.org/10.1016/j.neuroimage.2015.11.014.

[49] Slepian, D. Prolate spheroidal wave functions, Fourier analysis, and uncertainty—V: The discrete case. *Bell System Technical Journal* 57(5), 1371–1430 (1978). DOI: https://doi.org/10.1002/j.1538-7305.1978.tb02104.x.

[50] Thomson, D. J. Spectrum estimation and harmonic analysis. *Proceedings of the IEEE* 70(9), 1055–1096 (1982). DOI: https://doi.org/10.1109/PROC.1982.12433.

[51] Percival, D. B., and Walden, A. T. *Spectral Analysis for Physical Applications: Multitaper and Conventional Univariate Techniques*. Cambridge University Press (1993).

[52] Guerci, J. R. Artech House monograph (2003).

[53] Wicks, M. C., Rangaswamy, M., Adve, R., and Hale, T. B. *IEEE Signal Processing Magazine* 23(1), 51–65 (2006). DOI: https://doi.org/10.1109/MSP.2006.1593337.

[54] Melvin, W. L., and Guerci, J. R. *IEEE Transactions on Aerospace and Electronic Systems* 42(3), 983–996 (2006). DOI: https://doi.org/10.1109/TAES.2006.248215.

[55] Gurram, P. R., and Goodman, N. A. Spectral-domain covariance estimation with a priori knowledge. *IEEE Transactions on Aerospace and Electronic Systems* 42, 1010–1020 (2006). DOI: https://doi.org/10.1109/TAES.2006.248209.

[56] Van Veen, B. D., and Buckley, K. M. Beamforming: a versatile approach to spatial filtering. *IEEE ASSP Magazine* 5(2), 4–24 (1988). DOI: https://doi.org/10.1109/53.665.

[57] Kuglin, C. D., and Hines, D. C. The phase correlation image alignment method. In *Proceedings of the IEEE International Conference on Cybernetics and Society* (1975).

[58] Stone, H. S., Orchard, M. T., Chang, E.-C., and Martucci, S. A. A fast direct Fourier-based algorithm for subpixel registration of images. *IEEE Transactions on Geoscience and Remote Sensing* 39(10), 2235–2243 (2001). DOI: https://doi.org/10.1109/36.957286.

[59] Foroosh, H., Zerubia, J. B., and Berthod, M. Extension of phase correlation to subpixel registration. *IEEE Transactions on Image Processing* 11(3), 188–200 (2002). DOI: `https://doi.org/10.1109/83.988953`.

[60] Capon, J. High-resolution frequency-wavenumber spectrum analysis. *Proceedings of the IEEE* 57(8), 1408–1418 (1969).

[61] Frost, O. L. An algorithm for linearly constrained adaptive array processing. *Proceedings of the IEEE* 60(8), 926–935 (1972).

[62] Kay, S. M. *Fundamentals of Statistical Signal Processing, Volume II: Detection Theory.* Prentice Hall (1998).

[63] Scharf, L. L., and Friedlander, B. Matched subspace detectors. *IEEE Transactions on Signal Processing* 42(8), 2146–2157 (1994).

[64] Ward, J. MIT Lincoln Laboratory, Tech. Rep. 1015 (1994).